



**A SMART BILLING SYSTEM FOR ENHANCEMENT OF
RENEWABLE ENERGY TO NON-ELECTRIFIED
COMMUNITIES IN NIGERIA.**

BY

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ABSTRACT

There are many non-electrified communities in rural area of Nigeria, therefore many people cannot use electricity freely hence they result to alternative means, that is, running generators. Nigeria currently has high transmission and distribution losses that prevent all generated electricity from getting to consumers even though the country has the capacity to generate about 11kW, transmit 5kW, of which only about 3.5kW can be delivered to consumers. However, this transmitted power is not regular because of many reasons such as energy theft, vandalism of supply pipeline to the generating set, inadequate and poor infrastructure, increase in the population as well as an obsolete technology, hence inconsistency in billing results (estimated billing) which discourages new investment into the energy sector.

The objective of this research is to provide reliable billing system, whereby the gap between the current situation and the ideal situation of regular power and accurate billing is closed. That consumers can pay according to power usage using prepayment electricity metering system (smart billing technology).

This study focuses on a smart billing system (SBS) which achieves accurate billing directly related to exact energy consumed. This paper explores the design, implementation and application of SBS for electricity consumers in Nigeria. For low electricity consumption (domestic consumers), a smart prepaid meter has been designed which needs a prepaid card to keep the electricity supply continue. With SBS, this system can easily be enhanced to boast regular supply of electricity by encouraging investors in generation of renewable power plants (UNDP SDGs standard such as Solar, Wind, Hydro) to complement the existing plants hence enhancing more regular power supply.

The impact of this is that people can use electricity, pay as consumed and can also monitor and manage energy usage over a period of time as oppose to the current situation where bills are distributed indiscriminately to consumers. This solution eliminates estimated billing and bring about transparency to both the Distribution companies and the consumers.

Keywords: Pre-Paid, Post-Paid and estimated electricity billing system, electricity consumption, Subscriber satisfaction, Smart Billing System (SBS), Renewable Power Plants. Sustainable Development Goals (SDGs)

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All those who will read this document. I hope that it will be beneficial to you!

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Abbreviation

- ❖ KIC - Kobe Institute of Computing
- ❖ LED - Light Emitting Diode.
- ❖ GSM - Global System for Mobile
- ❖ Wi-Fi - Wireless Fidelity
- ❖ HTML - Hyper-Text Markup Language
- ❖ IoT - Internet of things
- ❖ NodeMCU - Open Source Microcontroller Unit
- ❖ API - Application programming interface
- ❖ NERC - Nigeria Electricity Regulatory Commission
- ❖ MDG- Millennium development goals
- ❖ SDGs - Sustainable development Goals
- ❖ PHCN - Power Holding Company of Nigeria
- ❖ GENCOs - Generating companies
- ❖ DISCO - Electricity Distribution Companies
- ❖ EEDC - Enugu Electricity Distribution Company
- ❖ EPBS – Electricity Payment Billing System
- ❖ MW - Megawatts
- ❖ TCN - Transmission Company of Nigeria
- ❖ SPMS – Smart Pre-paid Metering System
- ❖ IT - Information Technology
- ❖ ETAP - Electrical Transient Analyzer Program

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Chapter 1

1. Introduction

This chapter will focus on research area and describes the purpose of this master's thesis project.

1.1 Background

Nigeria is the largest economy in sub-Saharan Africa, but limitations in the power sector constrain her economic growth. Though the nation is endowed with large oil, gas, hydro and solar resource, and it already has the potential to generate 12,522 megawatts (MW) of electric power from existing plants, but most days is only able to generate around 4,000 MW, which is insufficient for the populace. The nation has privatized the energy generation and distribution section to accommodate investment in this section to allow increase energy production and better services to the people. This objective is far from being reached as the nation struggle with energy losses, theft, erratic power supply, irregular billings.

The advent of ICTs, their explosive expansion, and application to all sectors of activity brought a revolution and made ICTs tools we can reckon with when it comes to sustainable development many areas such as, government, economics, health, education, energy, transportation, communications, and agriculture. In energy, the positive impact of ICT is the improvement in generation, transmission and effective and efficient distribution to the consumers. The adoption of modern industrial inputs comes from ICT. KIC has one of the unique approaches in this regard and focuses on using ICT to solve social issues. Therefore, its motto is "Social innovation by ICT and yourself." Our research follows the same logic and tries to bring its contribution in this respect.

GENERATION CAPACITY

- Installed Capacity: 12,522 MW
- Thermal: 10,142 MW
- Hydro: 2,380 MW
- Reached Financial Close: 3,034 MW
- Power Africa 2030 Pipeline: 11,750 MW

CONNECTIONS

- Current Access Rate: 45%
- Rural: 36% Urban: 55%
- Households without Power: 20 million
- Target: Universal access by 2030
- Power Africa New Off-Grid Connections: 454,432

- Power Africa New Grid Connections: 496,723

INVESTMENT AND ENABLING ENVIRONMENT

Biggest Issues

1. Macroeconomic forces
2. Lack of creditworthy utilities
3. Lack of strong, transparent regulator

Power Africa Interventions

1. Loss reduction work with utilities
2. Transaction advisory services
3. Partnership with National Association of Regulatory Utility Commissioners.

POWER AFRICA'S ENGAGEMENT IN NIGERIA

MW GENERATION

Power Africa assisted the Government of Nigeria with agreements to move the Qua Iboe gas project closer to financial close. In parallel, Power Africa is assisting with agreements on several solar projects that will help Nigeria diversify its energy mix. Power Africa also helped Nigeria's first private IPP, the Azura Edo Project, reach financial close in 2015, including a \$50 million investment by The Overseas Private Investment Corporation (OPIC). The Azura plant became operational in 2018.[2]

CONNECTIONS

Power Africa, through the United States Agency for International Development (USAID) and the U.S. Trade and Development Agency (USTDA) is working to improve commercial operations and reduce losses at five distribution companies: Abuja, Benin, Eko, Ibadan, and Ikeja.

Power Africa is supporting off-grid options as well. With a \$15 million OPIC loan, Lumos, Inc. is deploying rooftop solar panel kits to approximately 70,000 residential and small commercial customers in Nigeria, using a lease-to-own business model. In partnership with General Electric, the U.S. African Development Foundation (USADF) and others, Power Africa has awarded nine \$100,000 grants to entrepreneurs for innovative, off-grid energy projects in Nigeria.

ENABLING ENVIRONMENT

Power Africa provides support to the Nigeria Electricity Regulatory Commission (NERC) through a partnership with the National Association of Regulatory Utility Commissioners (NARUC). This organization provides guidance on regulatory practices and tariff setting. Power Africa also provided planning support to the Transmission Company of Nigeria (TCN) to try to attract new investment in the transmission network.

SUCCESS STORY

Power Africa is assisting the Ministry of Petroleum Resources to structure and implement a program that will attract competent third-party off-takers to invest in the capture and utilization of gas flares using tested technologies. This work in support of the Nigeria Gas Flares Commercialization Program (NGFCP) will improve access to finance for gas flare projects to achieve quicker financial close through incentives to investors and strengthen capacity of regulatory agencies to monitor and sustain the implementation.

1.1.1 Tsuchida Lab

I conducted my research in Tsuchida-Lab under the supervision of Professor Masayuki TSUCHIDA. The laboratory is “Value Creation using ICT” meaning the approach of social innovation through existing ICT and new ICT technology that has values for customers and society. It is one of the labs at KIC, and it can collaborate with other labs of the school. Tsuchida-Lab focuses on value creation skill, ICT skill – basic ICT skills to make a prototype system, Business skill, - strategy, marketing, accounting etc and Human skill, - logical thinking, presentation, communication. It gives the student the opportunity to conduct his research by collaborating with a Japanese company if he so desires. Of course, he can do without it. Collaborating with a Japanese company allows discovering the Japanese working environment, exchange with the workers to know the challenges they are confronted to, focus on one of the challenges as a research topic and propose a suitable solution with ICT as the motto of the University advocates.

1.1.2 Eko Electricity Distribution Company (EKEDC)

As mentioned previously, I had the opportunity to collaborate with one of the companies for our research. That company, named Eko Electricity Distribution Company has agreed to work with us for the duration of our master course.

1.1.3 Company Profile

Eko Electricity Distribution Company (EKEDC) manages electricity in the southern part of Lagos State as well as Agbara in Ogun State. The new investors in the company are West Power and Gas Consortium. EKEDC has 11 Business Units which are Agbara, Ajele, Apapa, Festac, Ibeju, Ijora, Islands, Lekki, Mushin, Ojo and Orile and a growing customer population of 317,750.[3]

1.2 Problem Analysis

There are irregular or rather unfair billing system whereby about 50% of the entire population still receive estimated bills (No meter - fixed payment), about 41% on post-paid meter - analogue (use electricity before pay but hardly made payment) while about 9% have pre-paid meter (buying credits to load meter). The pre-paid meters display units bought or remaining and nothing more, it does not solve consumption recorded and remote monitoring from the distribution company hence energy can still be stolen bypassing the meter.

Nigeria electrification rate is 45% of Population

Economic Sector	GWh
Industry	4,057
Transport	-
Residential	14,003
Commercial and Public Services	6,379
Final Electricity Consumption	24,439
Losses	4,895

Source: IEA

Table 1: Electricity Consumption in 2014

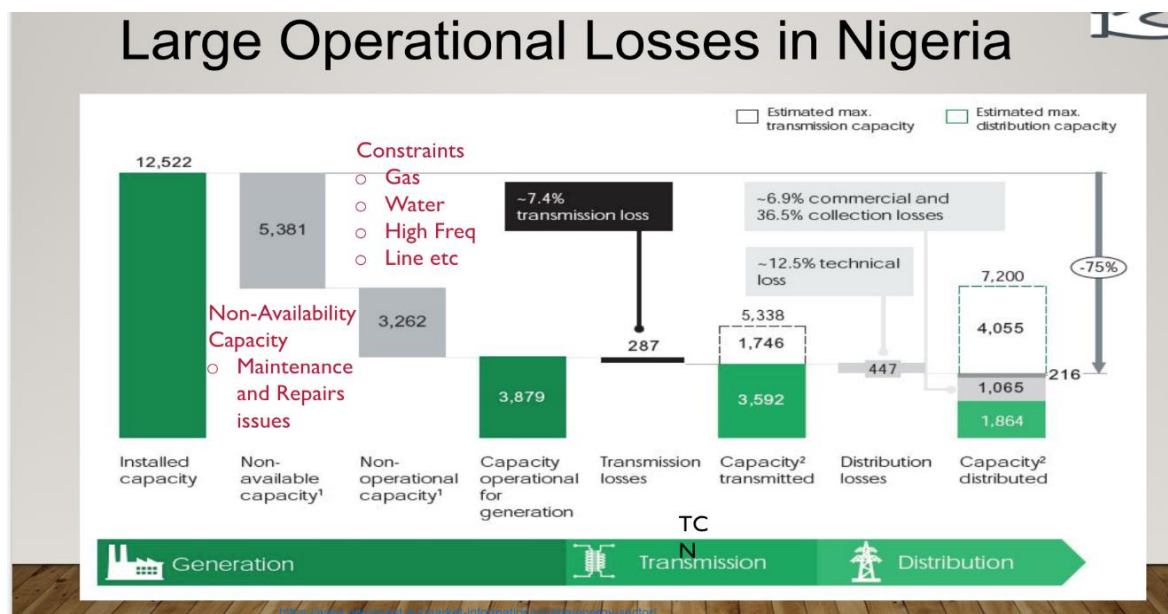


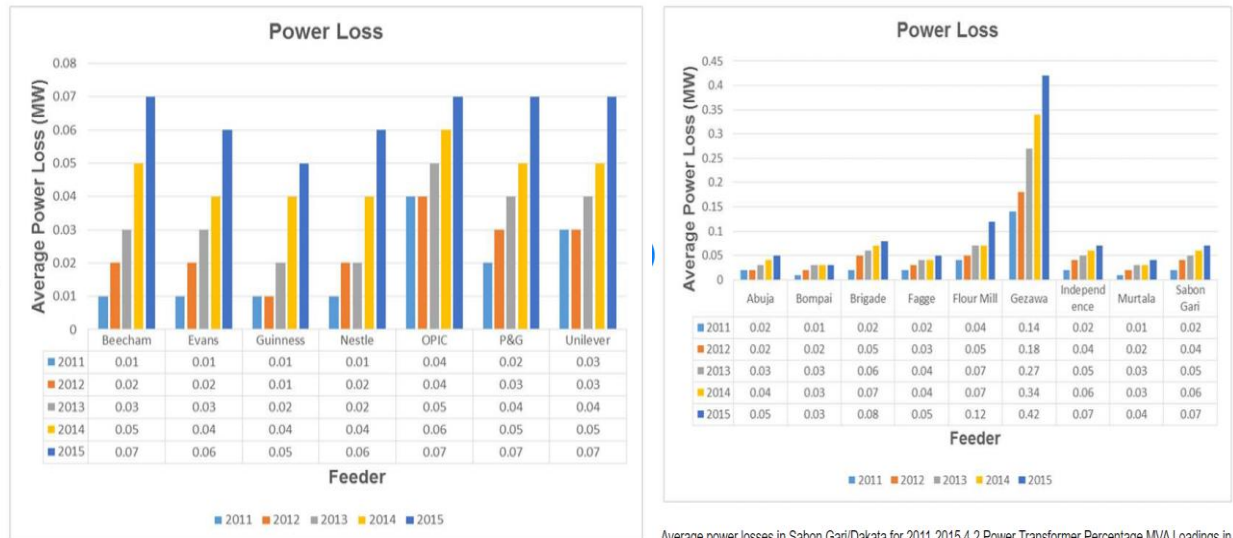
Figure 1: Electricity operational losses in Nigeria

1.2.1 Evaluation of Losses in Distribution Networks of Selected Industrial Cities in Nigeria Using ETAP

The number of power outages in a given locality per year is an indication of the efficiency of power supply even as consumer dissatisfaction with electricity service is often linked to high level of outages. Since the more the number of power outages, the less efficient the power system, it follows that given higher system losses (technical and non-technical), there would be more power outages indicating the inefficiency of the system

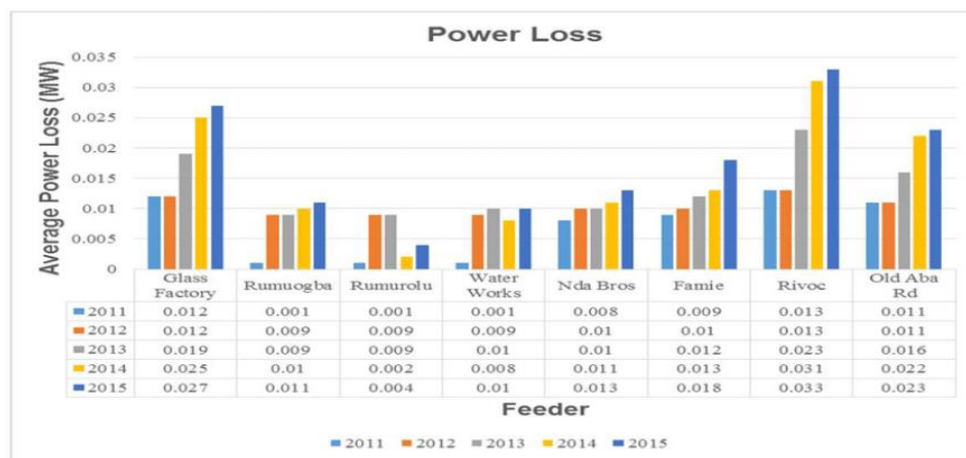
The biggest challenge confronting the Nigeria's distribution sector presently is the level of distribution losses namely technical, commercial and collection losses (Fig. 1). Technical losses are due to the nature of the components in use in the network, commercial losses refer to energy not billed for, and collection losses signify energy billed but not paid for. When the distribution

companies were privatized as part of the power sector reform process by the Nigeria government, the transactions assumed some particular loss levels. After the asset hand-over, however, it became clear that the losses were much higher than had been estimated. In 2014, ~46% of energy was lost through technical (12%), commercial (6%), and collection losses (28%) [1].



Average power losses in Agbara/Badagry for 2011-2015

Average power losses in Sabon Gari/Dakata for 2011-2015 4.2 Power Transformer Percentage MVA Loadings in the Distribution Networks



Average power losses in Akanni/Trans-Amadi for 2011-2015

Figure 2: Distribution losses evaluation in some selected cities in Nigeria

To solve this problem, many initiatives are being considered. Especially unfair billing prevents them from investing the power generation plants of renewable energy such as wind, solar, micro-hydro or biomass etc. in local communities in Nigeria.

Problem Statement

It is estimated that 4,740,275 electricity customers in Nigeria were unmetered as at February 2019. The DisCOS (Distribution companies) have funding or liquidity problems which they have largely

attributed to their inability to charge cost reflective tariffs and this has hindered their ability to provide meters.[4] They are also dealing with infrastructure challenges, energy theft and meter bypassing. In addition, the Multi-Year Tariff Order places some restrictions on the amount Discos can spend on capital expenditure.

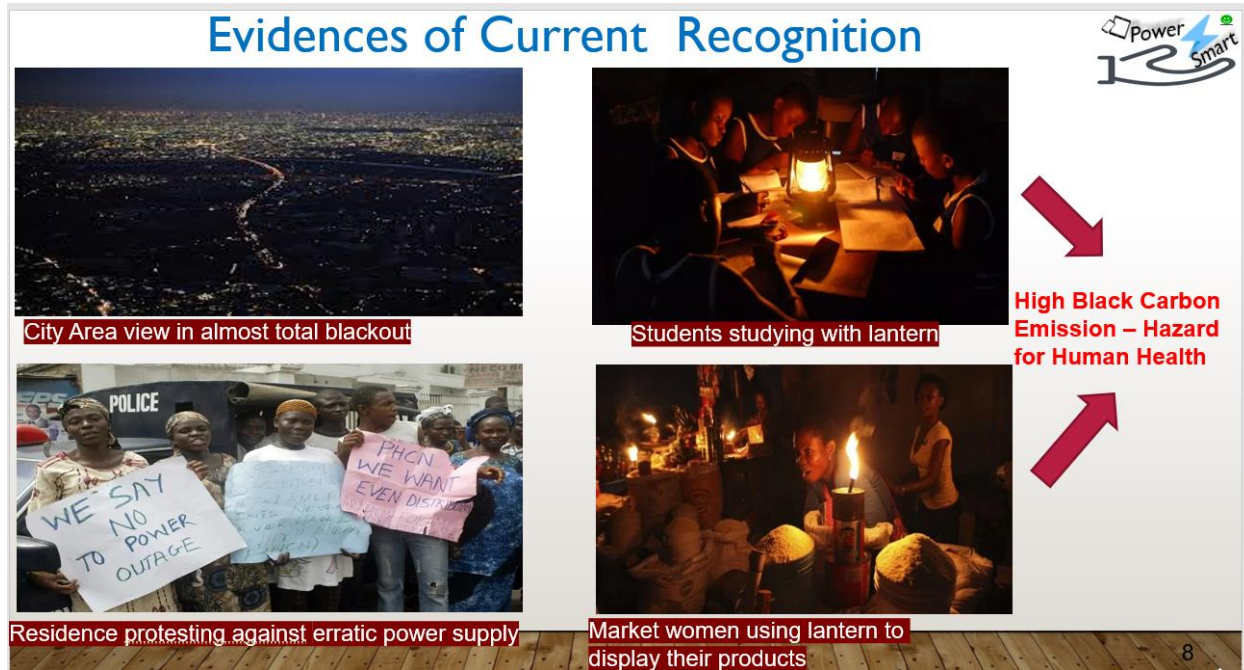


Figure 3: Present Electricity situation in Nigeria [5,6,7]

Eleven electricity distribution companies operate in the country and the grid is expanding to remote towns. But, more than 100 million Nigerians, out of a total population of 200 million, live in rural areas without access to electricity.

Abuja, Nigeria. May 7th, 2019 – The lower chamber of the National Assembly commenced a process to prohibit the issuance of estimated bills to consumers by electricity Distribution Companies (DisCos). The passage of the Bill, which scaled third reading on the floor of the Green Chamber, would ensure that electricity consumers must be provided with a prepaid meter, hence ending the estimated billing system and the system of paying for power not consumed. In line with the above, NOIPolls presents excerpts from the results and findings of its National Survey on prepaid meter ownership in Nigeria which was conducted in 2017. The survey revealed that ownership of fixed pre-paid metering is limited to only 9 percent of the population as the time this survey was conducted. According to the 2018 fourth quarter power report released by the National Bureau of Statistics (NBS); the report disclosed that the total number of electricity consumers with prepaid meters supplied by the 11 distribution companies (DisCos) in the country are 1,669,675.[8] More findings from the 2017 National Survey indicated that the rest of the population is divided among 41 percent who use post-paid meters and 50 percent who do not have a meter, but simply pay fixed amounts of money for electricity. Being the most efficient way to distribute electricity,

usage of pre-paid meters is highest in the South-East zone (13 percent) and lowest in the North-Central and North-West (7 percent apiece). Electric power usage in most cases is charged to non-urban users based on estimates that do not account for periods of power failure that are often prolonged. As expected, urban residents are twice as likely as rural residents to have pre-paid meters.

Which of the following applies to your household in terms of metering? (Disaggregated by Gender and Age-Group)								
Response	Nationwide	Gender		Age Group				
		Male	Female	18-35	36-60	61+		
Do not have a meter (Fixed payment)	50%	50%	48%	49%	49%	31%		
Have a post-paid meter (Use electricity before pay)	41%	41%	42%	41%	42%	58%		
Have a pre-paid meter (Buying credits to load meter)	9%	9%	10%	10%	9%	11%		

(Disaggregated by Geo-Political Zone and Urbanization)									
Response	Nationwide	Geo-Political Zone						Urbanization	
		North Central	North East	North West	South East	South South	South West	Urban	Rural
Do not have a meter (Fixed payment)	50%	45%	64%	69%	44%	43%	26%	45%	55%
Have a post-paid meter (Use electricity before pay)	41%	48%	28%	24%	43%	45%	63%	43%	39%
Have a pre-paid meter (Buying credits to load meter)	9%	7%	8%	7%	13%	12%	11%	12%	6%

Table 2: Electricity metering survey result in May 2019

Source:NOIPOLLS

In the Table 2 above, the survey has shown that access to prepaid electricity metering in Nigeria is too low, which directly affects the billings; therefore, further reforms of the electricity sector are imminent. It is essential to further improve accountability in the sector by subsidization of prepaid meters to enable low-income households acquire and use them as 50 percent of the population pay for electricity supply by estimated billing system which does not account for periods of power failure that are often prolonged.

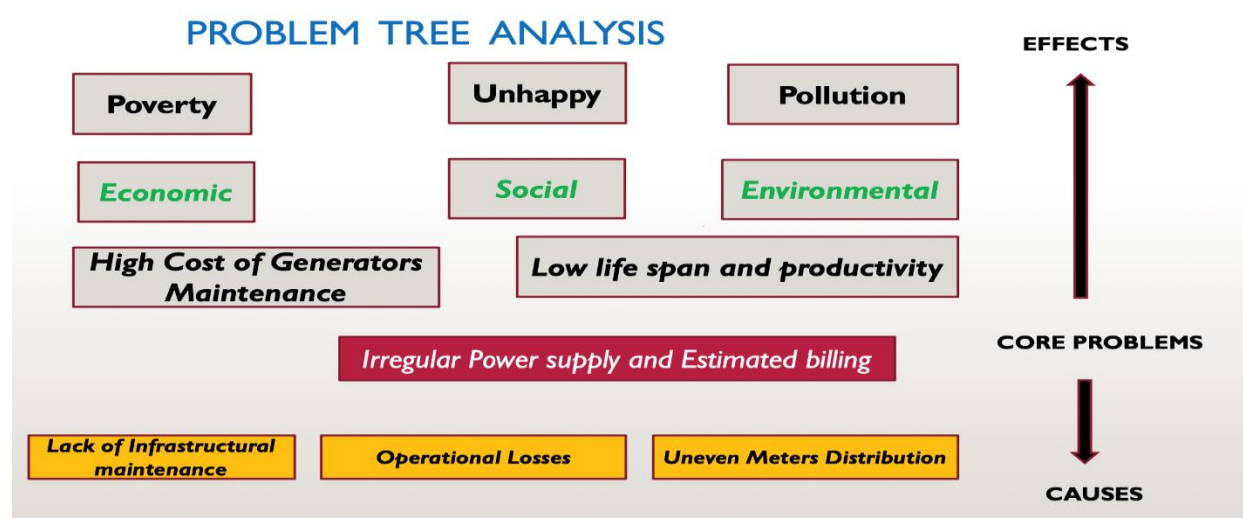


Figure 4: Problem tree/Analysis

- Close to 100 million, live in rural areas without access to electricity
- 50% of consumers are without electricity meter
- Low life span and productivity
- Unhappiness and Poverty

1.3 Hypothesis

- I will design and make an IoT device that makes the existing energy meter smart electricity meter design a prepaid smart meter for consumers without meters in order to tackle the poor metering issue that will allow the consumers to pay in advance accordingly to the quantity of electricity used.
- I will be able to integrate existing 6-digits token method of recharging pre-paid meter electricity purchase and online recharge.
- Input the 6-digits token on Electricity meter using keypad to charge your meter.
- When the units loaded expires the relay will cut-off and need to recharge again your meter message send to consumers.
- Customer will receive a friendly message before the units loaded expires for reminding to recharge again.

1.4 Purpose of The Study

The purpose of this research is to provide reliable billing system, whereby the gap between the current situation of unreliable billing, insecure network and the ideal situation of regular power, accurate billing and secure network is closed. That consumers can pay according to power usage using prepayment electricity metering system (smart billing system). That the network is secure by deploying appropriate affordable technology solution. This will to great extent will bring fairness in the part of the consumers and accountability and transparency on the distribution companies (DisCos). Also, the network will be secure and prevent any forms data theft resulting from cybercrime.

Therefore, investors could be encouraged to invest in clean energy (SDGs standard) sector of the country so that 50% of Nigerian in rural areas can have access to affordable regular or uninterruptible electricity supply.

Renewable energy is better (for the global environment, SDGs). In this context SDGs goal is readily applicable here which refers to insurance of access to affordable, reliable, sustainable and modern energy for all by year 2030.

1.5 Research Questions

The following are some of the research Questions developed to obtain data for this study:

- i. How can we improve electricity distribution in our communities?
- ii. What are the ways to eliminate estimated electricity billings?
- iii. How can we assist the distribution companies in detecting and resolving energy theft?
- iv. Release of CO₂ causes health hazard and global warming, how can we reduce or redirect this gas for effective usage (recycle for biomass energy)?
- v. How can the people of rural areas use electricity freely in Nigeria?
- vi. What are the major problems preventing wide coverage electricity in Nigeria?
- vii. What are the proposed innovative solutions to solve these problems?
- viii. What are government agencies doing about this issue?
- ix. What role is expected of private organization and investors in combatting these problems?
- x. What is ICT involvement in providing cost effective and affordable solution?
- xi. Nigeria is of the composition of 6-geopolitical zones, how will this solution affect all these zones without bias?
- xii. Would you like to buy your electricity pre-paid card via your mobile phone?
- xiii. Did you sometimes notice distortion in the line and irregular meter reading?
- xiv. Would you prefer a service that allows you to remotely monitor the Grid meters?
- xv. There is a huge loss (transmission, technical, commercial, collection etc) from Energy Generation to Distribution, will you prefer a system that automatically checkmate these losses and secure the network?

1.6: Research Approach

This research was conducted under Eko Electricity Distribution Company (EKEDC) online, one of 11 electricity distribution company approved by the government to interface with the consumers in Nigeria and its clients or populations, I have conducted interview with some relevant employee at EKEDC as well as with the electricity clients or consumers randomly. This interview was about the challenges both side EKEDC, and consumers were facing in terms of supply and payment respectively. Briefly I will do Quantitative research by using questionnaire and interviews where I will keep interviewing electricity distribution (EKEDC) and consumers and also Qualitative research by making a documentation or literature review in order to learn from the existing similar project and getting statistic numbers.

Development of IoT device to measure the energy consumption in a prepaid meter for consumer to monitor energy usage. And also, deployment of ICT solution for accurate billing and secured energy network.

1.7 Contribution

This project is intended to help electricity consumers have accurate electricity billings as against the present estimated, unmetered household and post payment system by introducing prepayment system. Having achieved aforementioned, the author believe that it will contribute to distribution of regular electricity supply which will enhance sustainable profit for the energy distribution companies (DisCOs) and consumers' benefit as well. The country will get increased revenues that

will help to improve electricity service delivery and the creation of job opportunity. Consumers will need electricity regularly for their personal business-like barbing, welding, electric car wash, etc). Adaptation on climate change effects that are causing global warming due to high rate of CO₂ generated by individual personal generators.

1.8. Thesis Structure

This thesis is comprised of 6 chapters. General introduction in this chapter. Chapter 2 describes the literature review, related work and the overseas research. Chapter 3, describes design and development, used components both hardware and software. Chapter 4 describes implementation and validation of the prototype. Chapter 5, conclusion and recommendation, in this chapter the thesis concludes with suggestions for future work and some reflections on issues that are relevant to this thesis project. Chapter6, appendix which includes prototype codes.

Chapter 2

2.0 Background Study

The demographic population is increasing exponentially in Nigeria hence the government of Nigeria has a vision to increase electricity generation, and distribution in order to satisfy the demand therefore they are trying to increase the generating plants in various district of the country and also keep searching the new technology to use for managing electricity to the end users which is the reason why a prepayment electricity meter was approved but the meters were not available to majority consumers, hence the estimated billings employed by DisCos to collect tariff. Smart prepaid system is still on agenda of the future projects to implement in Nigeria , therefore I am interested in making this research about IoT based prepayment electricity metering system in order to work together with government of Nigeria by providing some solutions and recommendation that will bring them to the implementation success of the new electricity meter system.

Nigeria is the largest economy in sub-Saharan Africa, but limitations in the power sector constrain growth. Nigeria is endowed with large oil, gas, hydro and solar resource, and it already has the potential to generate 12,522 megawatts (MW) of electric power from existing plants, but most days is only able to generate 3,500MW - 5,000 MW, which is insufficient for the growing population of over 200Million people. Nigeria has privatized its distribution companies, (DisCOs) so there is a wide range of tariffs depending on the location of this companies in the country.

2.1 Literature Review

2.1.1 The Implications of Estimated Billing System on Nigerian Electricity Consumers (Case study of one of the major Estates in Owerri, Imo State

This work takes a critical look at the electricity consumers consumption pattern with the use of estimated bills and prepaid meter bills in one of the major Estates in Owerri, Imo State, Nigeria, from two EEDC clients who are apparently living in the same building, but different flats. Their utility bills between the months of September 2016 and August 2017 were used in this study. The total load assessment for a three-bedroom home occupied by these consumers was carried out. A software-based graphical analysis of the data gotten from them was carried out. This analysis revealed implications of estimated billing system on these consumers as compared with the benefits of prepaid billing. This research was summed up with a far reaching recommendations which if implemented will curb the menace of estimated billing system in Nigeria

Power Holding Company of Nigeria (PHCN) was Nigeria electric power network operator and was over the years faced with the problem of revenue collection. This is mostly because electricity bills are sent to consumers after energy consumption. Consumers are reluctant to pay electricity bills due to the high estimated bills that is associated with the unreliable power supply. The low reliability of electric power supplies has little or no impact on the network operator because whether there is power or not, the normal estimated monthly electricity bills are sent to

consumers in the post-paid method. Therefore, the consumers suffer the cost of generating power for their individual usage(off-grid) and the cost of electricity that was never supplied by PHCN.

The defunct PHCN introduced the digital pre-paid meter in 2006 whose operation is like the loading of recharge card in the GSM handset. If power is available and the prepaid meter is loaded with units, the loaded units decreases only when load is connected and stops when power fails.

Generating companies (GENCOs) obtain funds by selling what they generate to DISCOs, since DisCOs are unable to realize all money for the energy sold to consumers, losses are incurred through power theft, hence GENCOs are underperforming in generation. DISCOs incur loss through three basic areas namely: huge unpaid bills by the consumers (**Collection loss**), power leakages due to poor, aged and insufficient generation, transmission and distribution network and power infrastructure (**technical loss**), illegal consumption of power by consumer who are customers of DISCOs and bypass of meters by consumer to reduce the power they pay for (**Commercial loss**).

The Aggregate technical, commercial and collection losses (ATC&C) in Nigeria is about 50% and as a result, there is high level of poor payment culture and this invariably leads to low power availability to the end users.

One of the great challenges faced by power supply in Nigeria from inception and was aggravated since handover from PHCN to private owners in 2013 is the controversy between the DISCOs and electricity consumers over the inappropriate billing system. There is arbitrary and lack of transparency in the method used by energy providers to cost and assess the customer energy consumption. Such dark practices manifest through estimated billing systems and irregularity in metering. It is noted that the bill for any current month is prepared before the end of the month, sometimes, indiscriminate charging due to loss of revenues incurred by DISCOs (Ofonyelu C.C and Eguabor R.E., 2014; Jain A, et al, 2011). The tariff system practiced in Nigeria for the electricity distribution, which is regulated by NERC, is called the Multi – Year Tariff Order (MYTO); a methodology to regulate fair electricity prices between consumers and DISCOs for efficient operation costs.

2.1.2 Multi Year Tariff Order (MYTO):

This is a tariff system designed for the Nigerian Electricity Market to provide a unified way to determine efficient total industry revenue requirement, and provide a 15-year tariff path in the power sector with minor reviews each year due to certain factors such as gas prices, exchange rate and inflation, and major reviews every five years, when all inputs are discussed with stakeholders. MYTO is used to set wholesale and retail prices in the Nigerian Electricity Market. On 18th December 2015, NERC, set a new tariff for DISCOs for 2015 to 2024, this new tariff requires consumers in different categories and location to pay different charges for each kWh (EEDC, 2014).

2.1.3 Factors That Determine Electricity Tariff In Nigeria

Formally, electricity tariff/charge depends on three elements, name, (1) Energy Charge (2) Demand Charge and (3) Fixed Charge,

Nowadays, the price for electricity depends on four major factors listed below:

1. Location; 2. Tariff Class; 3. Tariff Rate; 4. The quantity of consumed energy (in kWh).

1. Location

It is not a surprise that there are different electricity distribution companies (DISCOs) in Nigeria. They are all located at various regions of the country. There are 11 distribution companies and each of them covers a certain area, with varying tariff plan, namely;

1- Enugu Distribution Company; 2- Kano Distribution Company; 3- Yola Distribution Company; 4- Jos Distribution Company; 5- Kaduna Distribution Company; 6- Abuja Distribution Company; 7- Port-Harcourt Distribution Company; 8- Benin Distribution Company; 9- Ikeja Distribution Company; 10- Ibadan Distribution Company; 11- Eko Distribution Company.

2. Tariff Class

The amount of money you spend on electricity also depends on your tariff class. Electricity consumers in Nigeria are divided into 5 classes/categories, namely,

i. Class A - Residential: This category is strictly for residential energy consumers; R1 (lifeline), R2 (single and three phase), R3 (LV maximum demand), R4 (HV maximum demand)

ii. Class B - Commercial: This category is for consumers, whose premises are used for SMEs business e.g offices, beauty parlors etc. C1 (single and three phase), C2 (LV maximum demand), C3 (HV maximum demand).

iii. Class C - Industrial: This category uses their premises for manufacturing and other production processes eg. Shops and offices. D1 (single and three phase), D2 (LV maximum demand), D3 (HV maximum demand).

iv. Class D - Special tariff: These consumers include the following; religious houses, government buildings, educational establishments, hospitals, agro-allied industries. A1 (single and three phase), A2 (LV maximum demand), A3 (HV maximum demand).

v. Class E - Street lighting: This category includes S1 (single and three phase).

3. Tariff Rate and Quantity of Consumed Energy (kWh)

The Tariff Rate depends on your class. Main classes are divided by the DISCOs to sub-classes. The division is based on the average consumption of electricity by each class. It is different for each class and the Discos covering such area. It also depends on the quantity of consumed energy. Table 1 below gives the tariff for each DISCO and sub-class for the years 2017 and 2018 (2017/2018 in text). All numbers are in N/kWh.

Table 3. Tariff class of different DISCOs, showing the present and future tariff

S/N	Class	Enugu Disco Type – Amt. (₦)	Abuja Disco Type – Amt. (₦)	Ikeja Disco Type – Amt. (₦)
1 2 3 4 5	Class-A Residential	R1-4.00/ 4.00 R2S-30.93/31.00 R2T-34.28 /34.36 R3-48.12/48.23 R4-46.08/46.19	R1 - 4.00 / 4.00 R2 - 24.30 / 24.03 R3 - 47.09 / 45.72 R4 - 47.09 / 45.72	R1 - 4.00 / 4.00 R2SP - 21.10 / 18.94 R2TP - 21.73 / 21.15 R3 - 35.68 / 31.62 R4 - 36.11 / 32.00
6 7 8 9	Class –B Commercial	C1S - 34.28 / 34.36 C1T - 39.25 / 39.34 C2 - 45.24 / 45.35 C3 - 45.85 / 45.96	C1 - 37.39 / 36.25 C2 - 47.09 / 45.72 C3 - 47.09 / 45.72	C1SP - 25.56 / 21.76 C1TP - 27.59 / 24.45 C2 - 36.91 / 32.47 C3 - 37.30 / 32.81
10 11 12 13	Class – C Industrial	D1S - 40.37 / 40.46 D1T - 42.56 / 42.66 D2 - 45.67 / 45.77 D3 - 46.83 / 46.94	D1 - 36.07 / 34.96 D2 - 47.09 / 45.72 D3 - 47.09 / 45.72	D1 - 27.93 / 24.56 D2 - 37.54 / 33.02 D3 - 37.99 / 33.42
14 15 16 17	Class – D Special tariff	A1S - 34.16 / 32.24 A1T - 39.25 / 39.34 A2 - 45.26 / 45.36 A3 - 45.06 / 45.16	A1 - 35.74 / 34.63 A2 - 35.74 / 34.63 A3 - 35.74 / 34.63	A1 - 26.23 / 23.24 A2 - 29.53 / 26.17 A3 - 26.69 / 26.31
18	Class – E Street lighting	L1 - 32.86 / 32.94	S1 - 27.14 / 26.54	S1 - 18.99 / 16.83

This table shows that the numbers in each distribution company vary. However, some of the distribution companies clearly show an increase in electricity tariff for 2018. There is also the possibility of continued rise in the at least in the coming years.

2.1.4 Electricity Billing System In Nigeria

The DISCOs use a monthly billing system in which the unit that is consumed in a previous month is paid in the succeeding month. In a typical bill, there are eight columns which contain: Descriptions, tariff code, read date, present reading, previous reading, multiplier, consumption and current charges. Under the descriptions columns there are always two rows which are energy charges and fixed charges. There is a number in the consumption column which is calculated by subtracting previous reading from the present reading. However, sometimes a number with "E" will appear by the side under the consumption column like 150E. This means that the consumption figure is not gotten from the meter reading but an estimate figure. The current charge is obtained by multiplying the consumption figure with the energy multiplier charge, plus the VAT, which is dependent on the classification of the premises. For the electricity billing, Current charge = (consumption x energy multiplier charge) plus VAT in Nigeria.

Technically, electricity consumption is the amount of watts used over a period of time. So, a standard measure of electricity consumption is the amount of watts used over an hour: the watt-hour (Wh). For example, a 40W light bulb turned on for one hour will use 40Wh of electricity. An electricity bill records the number of kilowatt hours (kWh) consumed over a period, which is usually around a month when the meters are read. A kilowatt-hour is simply a thousand watt-hours. So, using a 4,000W air-conditioner for one hour will consume 4kWh of electricity (4,000 Wh) (Charpell, 2014). Example: For a consumer who uses 150 watts load per hour daily for one month, the Total Energy bill of the consumer will be calculated as follows if the per unit rate is N30.93k. (Take 1 month=30Days):

1 unit = 1kWh

Total kWh = 150 x 24 hrs x 30 days = 108,000 watts – hour

Converting this into units where 1 unit = 1kWh

Total Units Consumed; 108,000 / 1000 = 108 units

Cost per unit (Rate) is ₦30.93

Cost of Energy Consumed = 108 x 30.93 = ₦3,340.44

VAT → 5% of 3,340.44 = 167.02

Total due = 3,340.44 + 167.02 = ₦3,507.46

Therefore ₦3,507.46 is the total energy bill of the consumer for one month.

Table 4: Load Assessment of a Customer's Three-Bedroom Apartment at New Owerri, Imo State.

Type of Load	Power rating(W)	Quantity	Demand Factor	Actual Power(W)
Medium size deep freezer	130	1	0.5	65
Washing machine	280	1	0.5	144
Microwave Oven	1000	1	0.5	500
Electric Pressing Iron	1000	1	0.5	500
Air-Conditional	1170	1	1.0	1170
Refrigerator	500	1	0.5	250
Ceiling fan	100	5	0.7	350
Incandescent Bulb	60	23	0.7	966
LG 32" Television	100	1	0.5	50
Sharp 14" Television	80	1	0.5	40
Sony Home-theatre	100	1	0.5	50
DSTV Receiver	50	1	0.5	25
DVD Player	50	1	0.5	25
TOTAL				4135

Table 5; Estimated bill of an EEDC client at New Owerri District, Imo State for a year (Sept. 2016 to Aug. 2017) (EEDC Utility bill)

S/N	Month/ Year	Tariff Class (TC)	Rate (R) (₦)	Unit Consumed (UC) (kWh)	Energy consumed (EC)(₦)	VAT (5% of EC)	Energy due (ED) for the month (₦)
1	Sept. 2016	R2S	27.13	110	2984.30	149.22	3133.52
2	Oct. 2016	R2S	27.13	97	2631.61	131.58	2763.19
3	Nov. 2016	R2S	27.13	129	3499.77	174.99	3674.76
4	Dec. 2016	R2S	27.13	165	4476.45	223.87	4700.27
5	Jan. 2017	R2S	27.13	170	4612.10	230.61	4842
6	Feb. 2017	R2S	30.93	80	2474.4	123.75	2998.12
7	Mar. 2017	R2S	30.93	400	12372.00	618.60	12990.60
8	Apr. 2017	R2S	30.93	304	9402.72	470.14	9872.86
9	May 2017	R2S	30.93	188	5814.84	290.74	6105.58
10	Jun 2017	R2S	30.93	233	7206.69	360.33	7567.02
11	July 2017	R2S	30.93	146	4515.78	225.79	4741.57
12	Aug. 2017	R2S	30.93	146	4515.78	225.79	4741.57

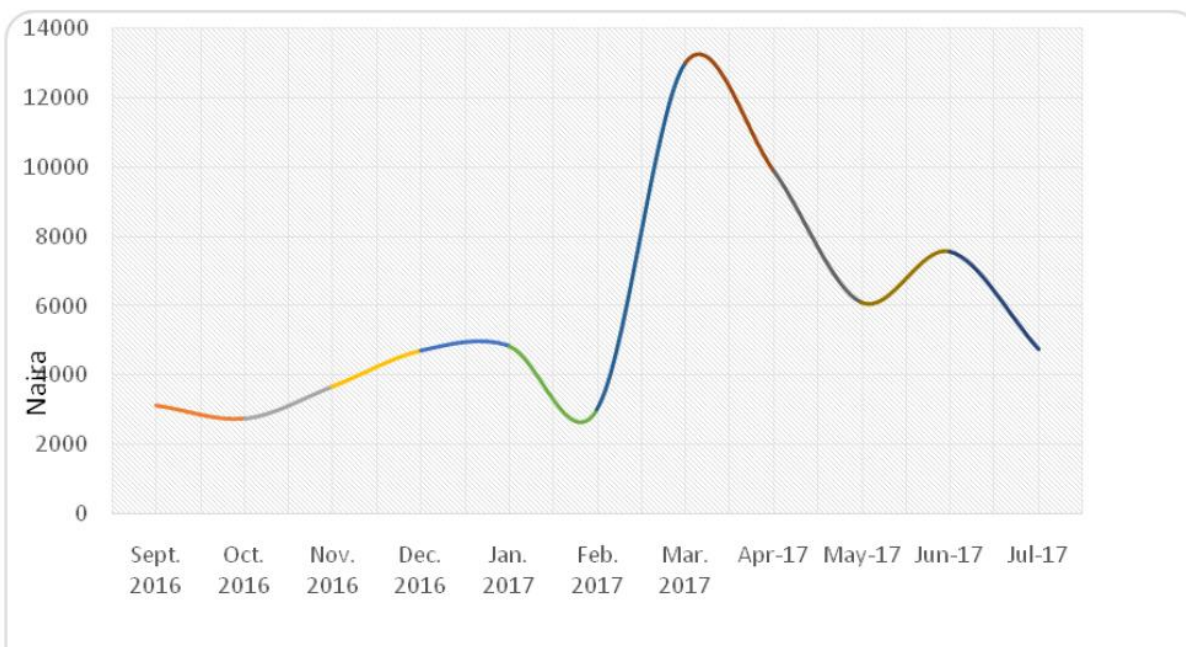


Fig 5: Graph showing the estimated bill of an EEDC client at New Owerri District for a year

S/N	Month/ Year	Tariff Class (TC)	Rate (R) (₦)	Unit Consumed (UC)(kWh)	Energy consumed (EC) (₦)	VAT (5% of EC)	Energy due (ED)for the month (₦)
1	Sept. 2016	R2S	27.13	95	2577.35	128.87	2706.22
2	Oct. 2016	R2S	27.13	97	2631.61	131.58	2763.13
3	Nov. 2016	R2S	27.13	105	2848.65	142.43	2991.06
4	Dec. 2016	R2S	27.13	102	2767.26	138.36	2905.62
5	Jan. 2017	R2S	27.13	105	2848.65	142.43	2991.06
6	Feb. 2017	R2S	30.93	50	1546.50	77.33	1623.83
7	Mar. 2017	R2S	30.93	50	1546.50	77.33	1623.83
8	Apr. 2017	R2S	30.93	105	2848.65	142.43	2991.06
9	May 2017	R2S	30.93	102	2767.26	138.36	2905.62
10	Jun 2017	R2S	30.93	85	2629.05	131.45	2760.50
11	July 2017	R2S	30.93	97	2631.61	131.58	2763.13
12	Aug. 2017	R2S	30.93	90	2783.70	139.19	2922.89

Table 6; Prepaid meter bill of an EEDC client at New Owerri (Area –A world-bank) District for a year (Sept. 2016 to Aug. 2017) (EEDC Utility bill)

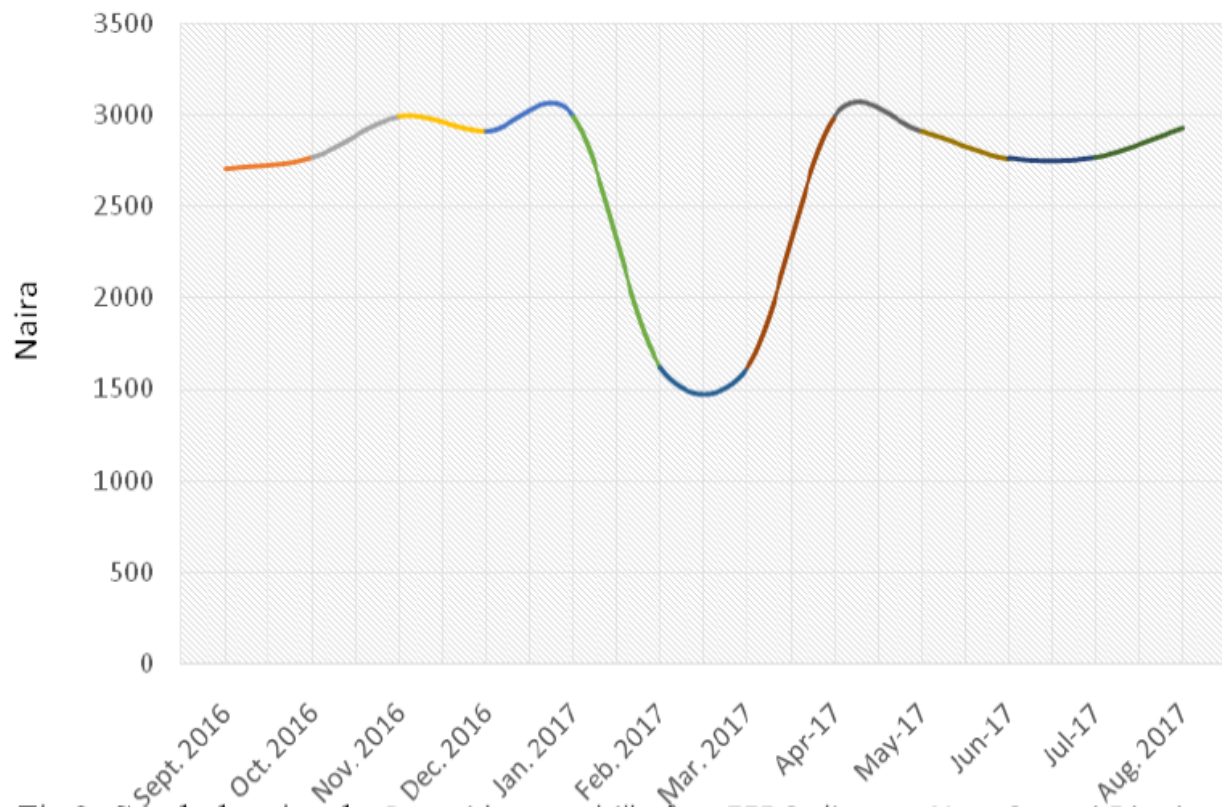


Fig 6: Graph showing the prepaid meter bill of an EEDC client at New Owerri District for a year

Tables 5 and 6 shows estimated bills and Prepaid meter bills respectively of two EEDC clients at Area - A, world-Bank housing estate New Owerri District for a year (Sept. 2016 to Aug. 2017), the two consumers are under the same service transformer and apparently living in the same building but different flats. This study painstakingly visited these clients and took the statistics of their energy consuming gadgets in their homes which were found to be the same. Client A is using the estimated billing system, while Client. B, uses the prepaid meter billing system. From the graph (curve) of energy consumed against the month/year, the highest energy consumed by B occurred in Nov. 2016, Jan. 2017 and April 2017. When he consumed N2848.65k. His consumption became lowest in Feb. and March 2017 which were rated at N1546.50k. This lower consumption was as a result of their faulty sub-station which occurred between 13th Feb. and 17th March 2017, hence making the consumption for that period to be low. On the average, Client. B consumes between 95 and 105 units per month which keeps the total energy consumed resonating between N2570, and N2850 per month.

For Client A who uses estimated billing system, the curve of Fig. 5 showed that he consumed the highest energy in March 2017 at N12,372 which apparently was the time their transformer had issue and could not supply adequately for the month. Also, his lowest consumption was noticed in Feb. at N2474.40 which was far above the lowest consumption for Client B which was at N1546.50 This study comparatively revealed that prepaid billing system is far better than estimated billing system because of the cost implications. Residential electricity consumption is

cheaper using the prepaid meter than the estimated billing system. This is obviously shown in table 6, Fig. 5 and table 4, Fig. 5 respectively

2.1.5 Implications Of Using Estimated/Unmetered Billing System

Having considered the available data and variables, it is obvious that estimated billing system has the following negative implications, thus;

- Over-billing of customers/ billing inaccuracies.
- Absence of service orientation to the customers.
- Collection of illegal money from the customers by the DISCOs officials.
- Illegal collection and stealing of power supply by the consumers.
- Inadequate power supply and metering system.
- Collection of disconnection / reconnection fees.

2.1.6 Benefits Of Using Prepaid Meter In Nigeria

Apart from the huge initial capital involved in supplying the meter to consumers by the utility companies, total replacement of estimated billing system with prepaid meters will guarantee the certainty that consumers will pay for the actual energy consumed as affirmed by Fagbohun O.O, et al. (2012) and Carolyne N. Metal (2013).

- Prepaid meter service creates opportunity of control of when and how much is paid for energy and effective usage. It helps in better budgeting.
- Reduces cost associated with billing, notification of disconnection, disconnection, and reconnection, customer service staff and call centers.
- Reduces bad debt and write-offs because arrears don't build up.
- Improves detection and management of outages.
- Metering and most importantly, prepaid system will enable the DISCOs to determine their revenues.
- Prepaid metering system reduces the level of contact between the energy provider officials and consumers,

2.1.7 Conclusion

Estimated billing system is said to be analogous to fraud where consumers are mandated to pay far above what they consumed monthly. Most of the customers under this system are without meters and their residences are never visited to track the energy utilization over the period charged. Provision of meters to individual customers may not totally eradicate this menace; however, it is a good step forward to the means. About 6 million electricity consumers in Nigeria were noted to have no meters to monitor their electricity consumption monthly. This encourages estimated and fixed charges allocated to the consumers which results in defrauding acts by the consumers not paying anything out of the over billed bills. For a better billing, energy conservation as well as need based energy management of resources, the prepaid energy metering of consumers' premises offers solutions to the long-experienced problems of inadequate supplies, corrupt acts been perpetrated by officials of utility companies at the consumer end. It ensures that the consumer pays only for energy consumed and not the estimated bill in postpaid billing systems which have been the bane of electrical energy consumers.

2.2 The Electricity Billing and Payment Process in Nigeria

Presently, electricity consumers in Nigeria can be broadly categorized into two: those without meters and those that have meters. The metered consumers are equally in two categories: post-paid meter users and pre-paid meter users. Therefore, three categories of energy users in Nigeria are: unmetered, prepaid and postpaid. Several efforts have been made to transform all electric users to pre-paid meter users. First, Nigerian Electricity Regulatory Commission (NERC) introduced the pre-paid metering system to electricity consumers in 2005, however, only few electricity customers got the pre-paid meter. Also, after the unbundling of the Power Holding Company of Nigeria (PHCN), the 11 succeeding private distribution companies have been making efforts to ensure that all electricity consumers are provided with pre-paid meters. Despite these efforts, the unmetered electricity consumers in Nigeria are about half of the registered electricity users as at 2014 (Oseni, 2015).

According to the Presidential Task Force on Power (2012), the proportion of the registered electricity users that are unmetered in Ibadan is 63%. The unmetered consumers are usually charged arbitrarily by electricity distribution companies. Such charge is against the ideal and global practice to charge unmetered customers based on the average consumption of all unmetered households in an area, which is often jettisoned by utility companies in Nigeria (Oseni, 2015). The energy policy initiatives to roll-out smart meters across the country is with the intension of altering consumers energy use behavior, with the ultimate aim of avoiding wastage and ensuring efficient use of energy. This is because, energy efficiency, especially in the household is influenced by the behavior of the household members, among other factors (Wilhite et al., 1996). Drawing implications from the billing system, therefore, enables the policy makers and energy users to identify energy-saving behavior and energy-wasting behavior (Ishak et al., 2012). Unmetered electricity consumers usually have unlimited consumption, the knowledge of which is capable of encouraging wastage. For instance, such consumers are not likely to switch of their bulbs, and other appliances even when they are not needed. It is a common practice to see electricity bulbs on in the afternoon in many places in Nigeria. The energy use behavior of the metered electricity Energy for Sustainable Development 40 (2017) International Energy Initiative. Published by Elsevier Inc. All rights reserved. Contents lists available at ScienceDirect Energy for Sustainable Development consumers (pre-paid and post-paid) is likely to be different, as they are likely to be mindful of their consumption.

The readings of some post-paid metered consumers are not taken regularly by the distribution companies. This could be as a result of laziness on the part of the electricity distribution companies' workers or schemes by electricity consumers. Some consumers are in habit of intentionally locating their meters out of the reach of the distribution companies' workers. The household energy use behavior of the post-paid metered consumers, whose meter readings are not taken regularly is likely to be inefficient. On the other hand, since pre-paid meters are designed to ensure effective increase in end-user involvement and engagement in energy saving, the energy use behavior of the pre-paid metered consumers is likely to be efficient.



Figure 7: Prepaid and Postpaid meters

2.3. The prepayment electricity metering related project

Electricity billing system

Several studies have analyzed electricity billing system in Africa and other continents. For instance, Tewari and Shah (2003) assessed South African prepaid electricity experiment. The study reviewed the economics, logistics, and technology underlying the South African experiment of prepaid electricity. The paper attributed the success of the program to good marketing campaign, innovative tariff schedules, better planning and management, and so on. Mwaura (2012) also analyzed the adoption of electricity prepayment billing system to reduce non-technical energy losses in Uganda, drawing lessons from Nigeria. The paper assessed potential benefits of the Electricity Payment Billing System (EPBS) in reducing power theft; understanding how EPBS operates and evaluating the possibility of EPBS adoption in Uganda. Findings indicated that for EPBS to be successfully adopted in Uganda, capital availability, proportion of EPBS targetable customers, energy use and revenue for those being targeted; and enforcement of a deterrent penalty for those apprehended stealing powers are to be taken seriously. Oseni (2015) examined the willingness to adopt prepayment metering (PPM) for a sample of Nigerian households that were not prepayment users. The estimated results revealed that decisions to adopt a prepayment meter are significantly affected by current electricity spending, current billing method and the split incentive problem. Whereas current electricity spending significantly increased the tendency to adopt PPM, the split incentive problem reduced the probability of adoption. Although unmetered consumers were more likely to express a willingness to adopt a PPM system than post-paid customers, they did not intend to pay a significantly higher amount to obtain the prepayment service. Income did not play a significant role in decision-making concerning PPM adoption and the corresponding WTP amount.

Also, outside Africa, Gans et al. (2013) estimated the effect of real-time usage information on residential electricity consumption in Northern Ireland. They relied on this event that account

for the endogeneity of price and payment plan with consumption through a plan selection correction term and found that the provision of information is associated with a decline in electricity consumption of 11–17%. They also found that the reduction is robust to different specifications, selection-bias correction methods and subsamples of the original data. The advanced metering program delivers reasonably cost-effective reductions in carbon dioxide emissions, even under the most conservative usage reduction scenarios. Chou and Yutami (2014) analyzed smart meter adoption and deployment strategy for residential buildings in Indonesia. For countries pursuing sustainable development and energy efficiency, the use of smart meters is considered a first step in allowing residential consumers to remotely control their energy consumption, and a promising. The study enhanced the understanding of consumer perceptions and behaviors and can help decision makers and energy utility companies develop policies and strategies for a “one-size-fits-all” program related to smart meter applications in future residential buildings.

Torriti (2014) compared the sustainability impacts of smart meters and load controllers in an occupied office building in Italy. Government initiatives in several developed and developing countries to roll-out smart meters call for research on the sustainability impacts of these devices. Findings showed that demand reductions associated with a smart meter device are 5.2% higher than demand reductions associated with the load controller. O'Sullivan et al. (2014) reported on a longitudinal interview study of consumers, who were either using prepayment metering or having trouble paying their electricity bills to explore how prepayment metering influences household budgeting and management of electricity use. The study highlighted that better regulation of the presently market-led electricity prepayment metering systems used in New Zealand could reduce the disadvantages while capturing the potential benefits of using prepayment metering for consumers.

Household energy use behavior

Energy use behavioral models are commonly deliberated in economic psychology. In the context of energy consumption, the consumption of energy is not behavior but rather it is the consequence of behavior (Martiskainen, 2007). Such consequential behavior includes turning the lights off or lowering thermostat levels (Becker et al., 1981). Becker et al. (1981) analyzed the behavior which relate to households' direct energy requirements (electricity and space heating), including behaviors such as turning lights on, using electric appliances, adjusting thermostat settings, cooking and washing. The study also briefly discusses sustainable consumption behaviors, which are closely linked to purchasing decisions such as the buying of energy efficient appliances. Wilhite et al. (1996) compared and contrasted the results of ethnographic investigations of energy use behavior in Fukuoka, Japan

In addition, this work will open access to clean modern energy services by investors which is an enormous challenge facing the African continent because energy is fundamental for socioeconomic development and poverty eradication. As such, 60% to 70% of the Nigerian population that does

not have access to electricity will be provided with regular power, accurate billing and secured network with the adoption of this technologies to reduce energy wastages and to save cost.

2.4 Related Work

In 2010, linear detection algorithm is used to detect which advances are active in their power contributions but due to the errors in the data base it contains many problems. Problems are robust to errors in this database. In the adjacent year 2011 using cloud computing technology found the solution for efficiency calculation of individual equipment [9]. Later In 2012, using three feedback system, monitored the energy in residential Real-Time. But there is a problem when we try to save the energy due to continuous engagement of the device. During the 30-days after installation residences result in low residential consumption rates which is used to determine the feedback provided by the Real-time energy monitors [10]. In 2013 a smallest ZigBee compatible node in existence came into picture which is known as green technology. This technology will possible in every place sensing different data types from energy metering to the environmental monitoring. Eventually 1n the next year 2014, GSM technology implemented automatic power will be reading. In 2016, Using Wi-Fi technology application can develop for Apple and BlackBerry 10 OS, thus providing multiple platform users support [11]. In 2017, using IoT technology An IoT device was created which measures the voltage, current, power and energy of 3 phase power line

Chapter 3

3.0 System design and Development

In this chapter I described the technologies used to make my electricity meter devise and other technologies for the whole project, the necessary items, and the architecture that have been used to develop our model system. In this chapter we described the prototypes for a system to monitor the Electricity consumption payment in Nigeria by using Internet of Things technology as followed below.

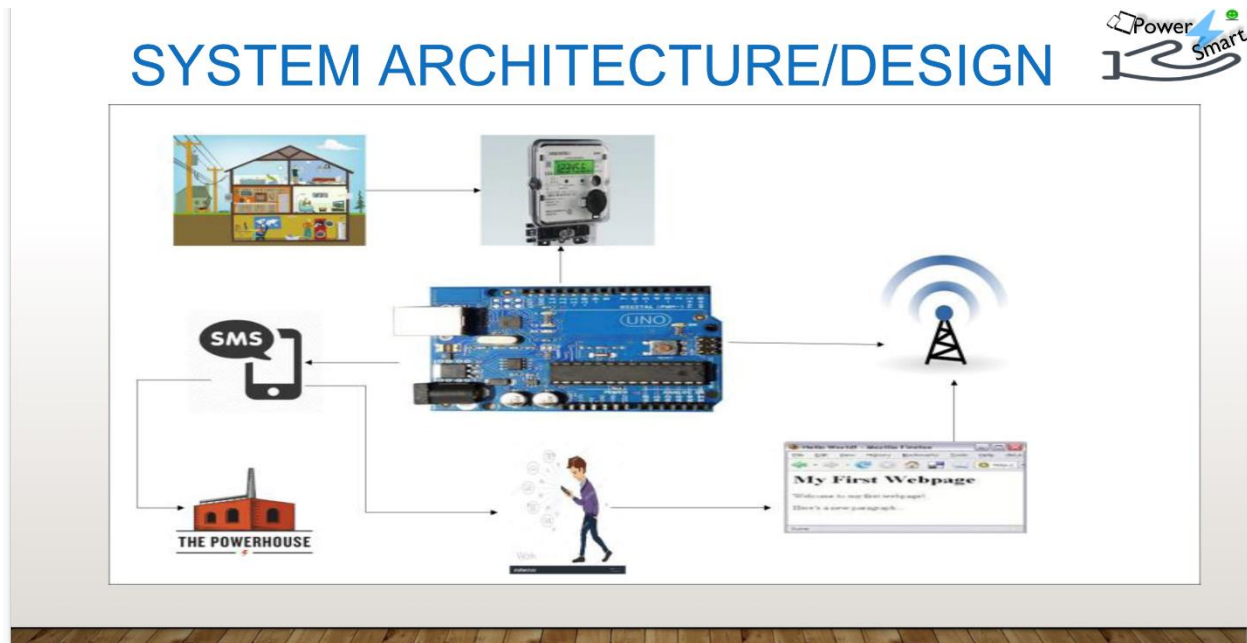


Fig 8: consumer services design

- ❖ The customer will pay electricity in advance and consume after.
- ❖ When the electricity credit expires relay will cut off supply so that a customer must recharge again his/her electricity meter by buying other electricity credits.
- ❖ The message alert will be provided to the customer to inform him that his electricity credit is below or not enough and reminding him to recharge again before time. We do not want to see our customers getting surprised by seeing electricity cutting off suddenly, therefore that message will be provided for a friendly remind to recharge electricity meter in early time.

3.1: Consumers Services Experience Map

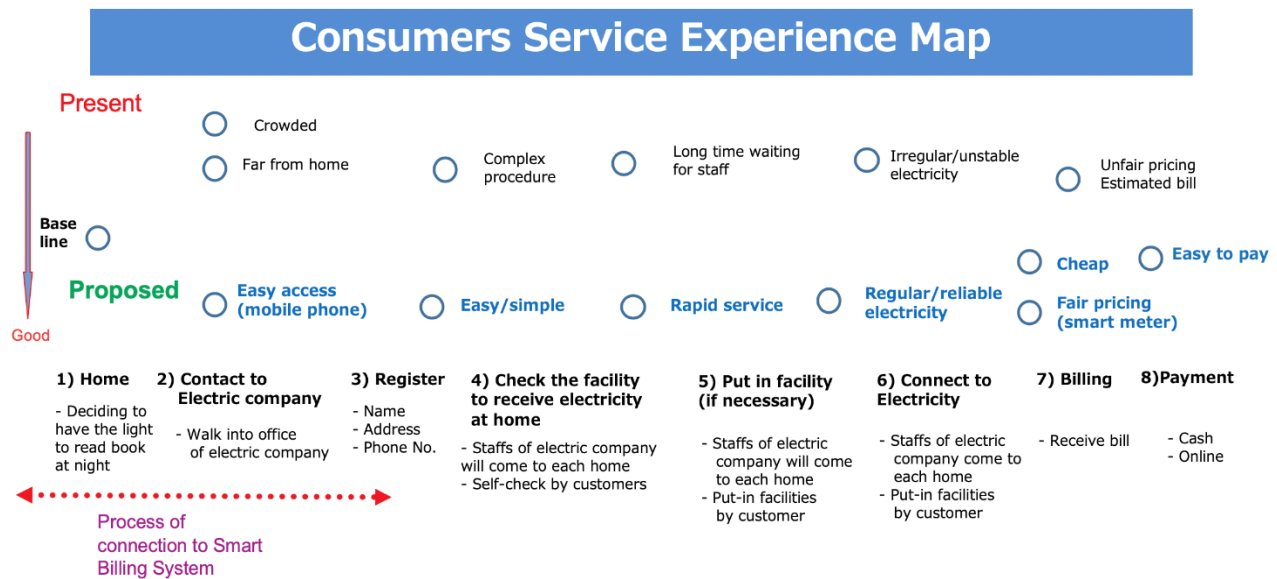


Fig 9: consumer services experience

Consumers Expectation

- Consumers are expected to register with energy company (DisCo) in their district so as to have their details like names, address, meter number, telephone number etc stored in the database.
- Enable consumers have access to new facilities offered by the new system.
- Atleast one member of the household should have a mobile phone to receive sms message in the least case.

3.1.1 System Requirements

The main idea was to build affordable system that could help to improve electricity management and payment in Nigeria using affordable technology, I have selected the best micro-controller and all running on Arduino IDE reading different type of sensors and reporting to Message Queuing Telemetry Transport(MQTT) and thing Speak cloud platforms.

3.2. Hardware Specification

In prototyping of my electricity meter project, I have used different hardware items and I have chosen some of them or the main hardware items to be explained more about them. Below are some hardware items that has been used in my prototyping devise:

- ACS712 Hall Effect 30Amp current Sensor
- Arduino Uno
- Energy Meter / AC Appliances
- ESP32/NodeMCU – ESP8266 – Arduino Wemos

- Male – Female Wires
- 16 x 2 LCD Display
- Breadboard

Below is the explanation of the main hardware items such as Arduino Uno Wi-Fi, Electricity Flow Sensor and Plastic solenoid valve.

3.2.1 Arduino UNO and ESP8266 Wi-Fi

In my prototype Arduino Uno and ESP8266 Wi-Fi was used as an access point to broadcast internet and get to send to our cloud database. Arduino Uno is a fully functional computer, it is faster when it comes to clock speed, it can multitask, support one USB port and connect through wireless to the internet. In my project prototyping, I have used the Arduino Uno that has the following specifications:

- 8-bit Microprocessor from Microchip
- Provides 2.4GHz, 802.11 b/g/n Wi-Fi
- ECC608 Cryptographic Chip Accelerator
- Secure Storage of 16 Keys, Certificates or Data

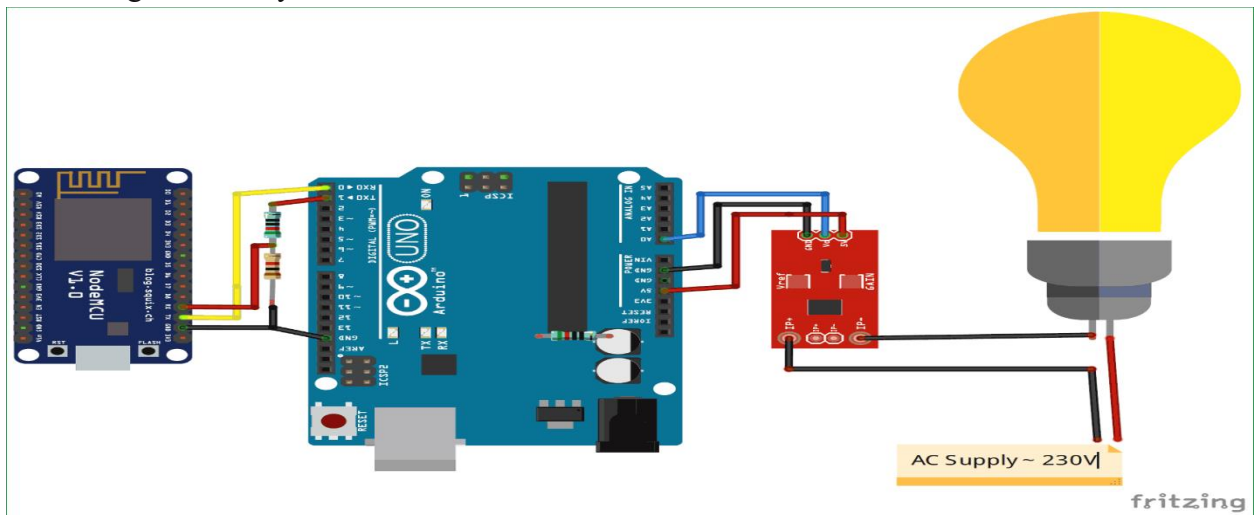


Figure 10: Arduino UNO with current sensor ACS712 and ESP8266

3.2.2. Current Sensor

In this project I have interfaced ACS712 current Sensor with Arduino to measure energy consumption. This sensor sits in line with ESP12 WiFi module to measure how much current and voltage consumed through the Arduino transfer the data wirelessly on the cloud for consumers to monitor on their mobile devices. There's an integrated magnetic hall effect sensor that outputs an

electrical pulse with every revolution.

- ◆ Hall Effect Sensors are transducer type components that can convert magnetic information into electrical signals for subsequent electronic circuit processing.
- ◆ Current sensors use the Hall Effect to convert current inputs into voltage outputs.
- ◆ In the Hall effect, electrons from an electric current flow through a magnetic field plate.
- ◆ The field then causes the electrons to "push" to one side of the plate and produce a voltage difference between the two sides.
- ◆ The difference in voltage from the side of the plate is the output of the sensor.

This sensor was used to measure liquid flowing through a pipe or container, or to create a control system based on the Electricity flow rate or quantity. For example, you could use this sensor or system at your home to measure the amount of electricity used at home in order to prevent energy wastage. I have to know the quantity of electricity consumed before to charge the payment from customer, that is the reason why I choose to use this sensor so that I can know the amount of electricity consumed by each customer. This sensor has a capacity to measure energy that is being consumed.

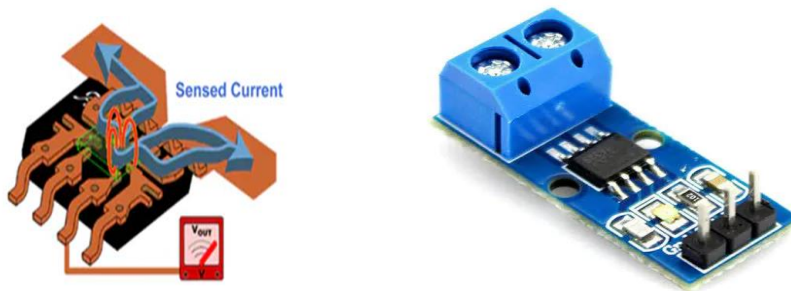


Fig11: The working principle of hall effect of ACS712 Current Sensor

3.2.2.1. Features of Current Sensor

Table 7: Sensor Specifications

	5A Module	20A Module	30A Module
Supply Voltage (VCC)	5Vdc Nominal	5Vdc Nominal	5Vdc Nominal
Measurement Range	-5 to +5 Amps	-20 to +20 Amps	-30 to +30 Amps
Voltage at 0A	VCC/2(nominally 2.5Vdc)	VCC/2(nominally 2.5Vdc)	VCC/2(nominally 2.5Vdc)
Scale Factor	185 mV per Amp	100 mV per Amp	66 mV per Amp
Chip	ACS712ELC-05A	ACS712ELC-10A	ACS712ELC-30A

- ◆ We test the sensor by giving 12 measurement points from -2A to 2A input range.
- The voltage reading of each corresponding measurement is seen as:

Input Current (A)	Output Voltage (V)
-2	2.148
-1.5	2.225
-1	2.312
-0.5	2.403
-0.2	2.44
0	2.474
0.2	2.505
0.5	2.546
1	2.645
1.5	2.727
2	2.817

Table 8: Output voltage reading of ACS712 current sensor

Output voltage reading of ACS712 current sensor Plotting of the obtained data Regression of voltage versus current

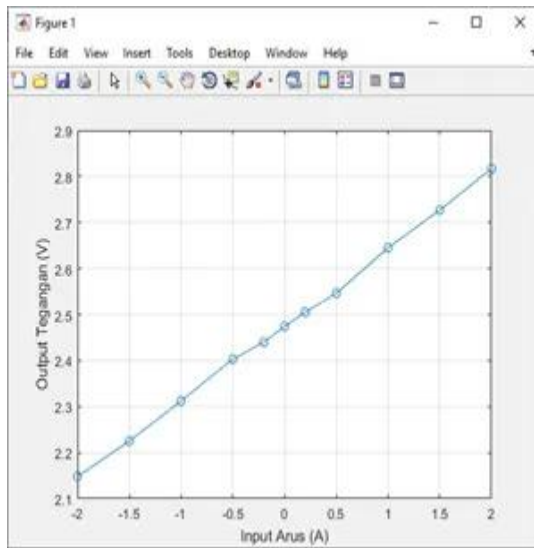


Fig 12: Regression of the obtained result shows the relationship

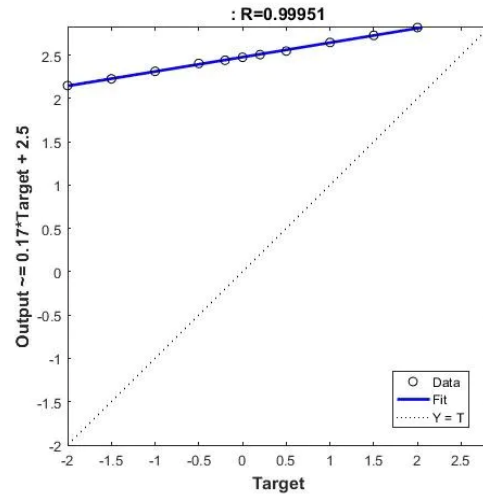


Fig 13: Regression of voltage versus current

$$V_{out} \approx 0.17I_{in} + 2.5 V$$

This result shows a 2.5V output at 0A and a 170mV/A slope as expected from the reference sheet

3.2.2.2. The Functionality of the Hall Effect Current Sensor:

In my prototyping of IoT based prepayment electricity metering devise, I used this sensor to measure the current & voltage by using Arduino Wi-Fi works on the principle of the Hall effect. According to the Hall effect, a voltage difference is induced in a conductor transverse to the electric current and the magnetic field perpendicular to it. Here, the Hall effect is utilized in the current meter using by providing isolation between the load (AC/DC load) and measuring unit (Microcontroller part), which is placed in the path of the current flowing.

For this project, we use a power supply (giving output voltage from 0 to 5 Volt) and a 1 ohm resistor 8 watt to generate current from 0 to 5 Ampere. Later on, we'll measure the voltage output from the ACS712 Current Sensor.

We use Arduino UNO to give ACS712 5V power supply (to 5V pin in ACS712). The ACS712 sensor's ground is also connected to the Arduino UNO ground.

For measurement, we connect the + probe of voltmeter to analog output pin in the ACS712.

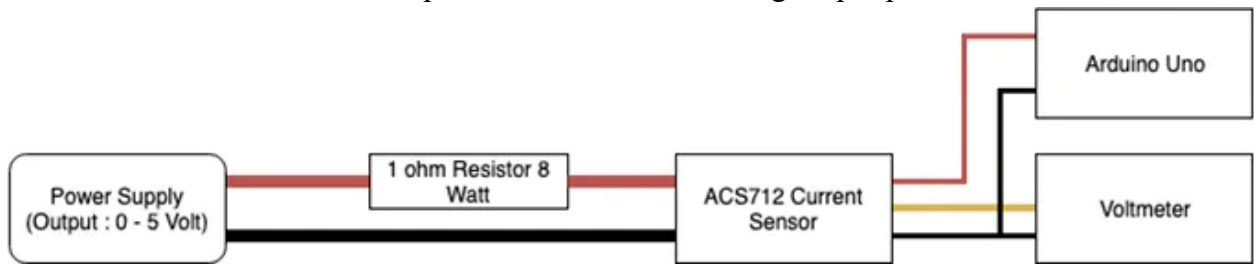


Fig 14: Schematic for measurement

About NodeMCU ESP-12

The specification of this module

- ISM Provides 2.4GHz, 802.11 b/g/n Wi-Fi
- PA +25dBm
- 8-bit Microprocessor from Microchip
- Secure Storage of 16 Keys, Certificates or Data

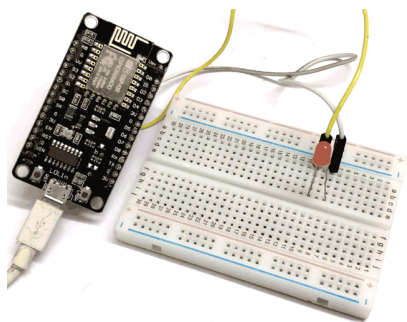


Fig 15: Connecting ESP12 on breadboard.



Fig 16: ESP8266 WiFi Module

Most people call ESP8266 as a WIFI module, but it is actually a microcontroller. ESP8266 is the name of the microcontroller developed by Espressif Systems which is a company based out of shanghai. This microcontroller can perform WIFI related activities hence **it is widely used as a WIFI module.**

There are many types of ESP8266 module available ranging from ESP8266-01 to ESP8266-12. The one that we are using in the tutorial is the **ESP8266-12.**

However, all the ESP modules have only one type of ESP processor, what differs is only the type of breakout board used. The breakout board of ESP8266-01 will have only 2 GPIO pins whereas ESP-12 has 16 GPIO Pins.

Many equipment's around us cannot be connected to internet on its own. So, this chip enables this equipment to connect with internet and make them IoT devices. Now, you can easily open your doors for IoT Projects with the help of this module. This low cost, small size module could do wonders and is really simple and easy to use, provided we follow the right steps.

NodeMCU is a Firmware on ESP8266. It is **ESP-12** and it is basically a System on Chip (SoC).It has Lua based firmware which is used to develop IoT based applications.

Posting Data to Google Sheets Using ESP8266

In the past couple years, the ESP8266 platform has flourished dramatically and emerged as one of the most popular hardware tools among electronics hobbyists and IoT enthusiasts. Packed with a 32-bit RISC CPU running at 80 MHz, a fully integrated Wi-Fi radio with TCP/IP protocol stack, serial peripherals (I2C, SPI, and UART), an ADC channel, and general purpose I/O pins, the ESP8266 is the most integrated and affordable Wi-Fi solution available in the current IoT market space. An ESP8266 hardware, like NodeMCU and ESP-01, can directly interface with sensors using its peripherals and upload the sensor measurements to a local or a remote web server via internet. Right now, there are already quite a bit of cloud IoT platforms (ThingSpeak, thinger.io, TESPA.io, Xively, ... the list is getting bigger everyday) that provides APIs and tools to allow the ESP8266 users to directly upload their sensor readings online for real-time visualization and global access. If you are a regular user of Google Drive, like me, you would find a Google sheet more approachable than all those IoT cloud platforms. In this tutorial, I will describe a method of connecting the ESP8266 device directly to a Google sheet for storing the sensor data without using any third-party plugin.

Basically, there are two parts to this project. The first part is setting up the ESP8266 hardware to read the soil moisture sensor output. The second part involves creating a Google sheet and configuring it to accept the sensor data sent by the ESP8266 module over the internet using a Google App Script that is provided with this project.

Hardware setup

This project uses a very simple and inexpensive current sensor consisting of three pins on the circuit board. The accompanying instrumentation board provides an analog output voltage that varies with the voltage level of the measured current. The instrumentation circuit is powered with a 3.3V supply and its output is fed to the analog input channel (A0) of the NodeMCU.

The ESP8266 analog input can handle the maximum input voltage of 1V. However, the NodeMCU implements a voltage divider circuit on board to expand the range of input voltage to 3.3V.

Software

We obviously be sending data using a GET request over an URL which has a pattern of **https://script.google.com**. When you enter this URL in a web browser, the Google's server responds back asking the browser to redirect to another URL with domain **script.googleusercontent.com** with a new GET request. For a web browser, the URL redirection is a very common thing and it works without any problem. However, for ESP8266, this is not straight forward. The ESP8266 needs to correctly decode the header information received from the first server to extract the redirect URL and make a second GET request to the new server. To make this redirection simpler a shared piece of code on GitHub in the form of Arduino Library named **HTTPSRedirect**. There are other examples of Arduino posting data to Google

spreadsheets using a third party API service, like **pushingbox**, for handling Google's https requirement and URL redirection. The use of **HTTPSRedirect** library makes the task much simpler by avoiding the need of any third party service. So, the first thing you need to do is to copy the HTTPSRedirect library files from GitHub and install into your Arduino libraries folder.

Program ESP8266 to send data to Google Sheets

Posted below is the ESP8266 code written using Arduino IDE for reading the soil moisture sensor output and posting it to a Google Sheet on your Google Drive. In order to make it work for your case, you will need to update the **SSID** and **password** matching to your Wi-Fi network. Also, you will also need to get the ***GScriptId**, which can only be obtained after publishing the required Google App Scripts. The process of obtaining the ***GScriptId** is explained towards the bottom of this tutorial. The sensor data are posted to the Google sheet every 15 minutes.

3.3. System Architecture

The proposed system architecture is composed with the low-cost environmental IoT devices that are connected to the cloud platform. In this research, I have used Arduino Uno with Wi-Fi module that can transmit data to the nearby access point and upload data to cloud platform (Thing-speaks). And I have chosen to use LoRaWAN as IoT network connectivity because of the great specifications as followed below:

- LoRaWAN (Long Range Wide Area Network) is a protocol for wide area networks. It is designed to support huge networks (e.g. smart cities) with millions of low-power devices.
- LoRaWAN can provide low-cost mobile and secure bidirectional communication in various industries.
- Standard: LoRaWAN
- Frequency: Various
- Range: 2-5km (urban area), 15km (suburban area)
- Data Rates: 0.3-50 kbps

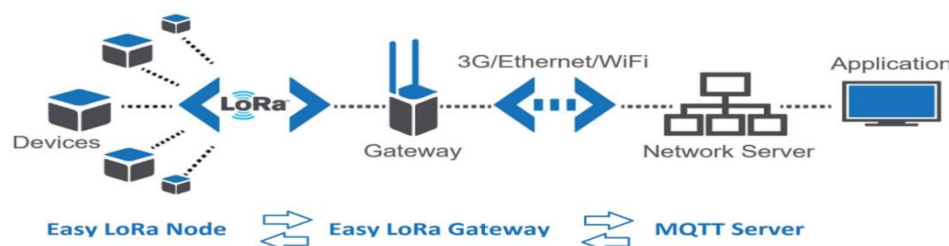


Fig 17: System architecture of the proposed electricity billing system

As it is said above that is IoT based project, I have used MQTT data Protocol to send data to the cloud platform in case of insufficient Electricity, the warning message function is optional in my smart electricity meter.

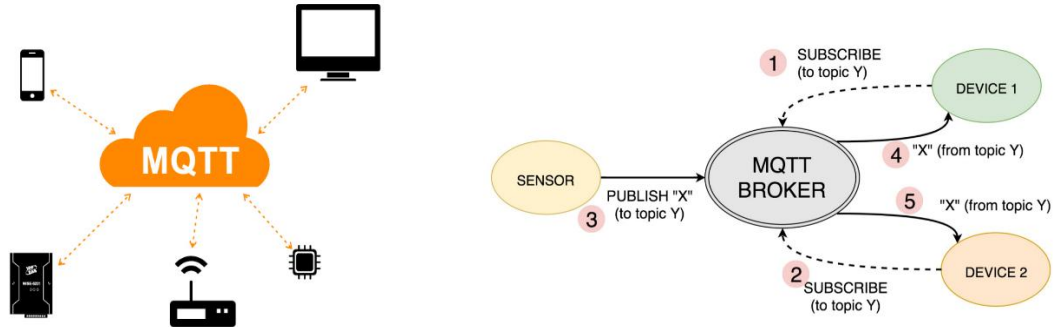


Fig 18: Message Queuing Telemetry Transport Protocol

3.4. The proposed System

I have tested my device, the results proved that this system will be used to collect Electricity revenue based on the consumption and will help customer to monitor their daily, weekly or monthly electricity consumption as the case may be; and know how to manage electricity by themselves. Even though it still has much to be done but I proposed, and I am expecting my system to work as followed below:

- My System, in order to tackle the poor metering issue, it allows the consumer to pay in advance accordingly to the quantity of electricity they need.
- Consumer can buy from existing approved vendors a recharge energy card
- Allow consumer to load the digit into prepaid smart electricity meter using keypad to charge your meter or online through mobile device.
- When the unit in the card expires the relay has to cut-off and need to recharge again your meter message is sent prior to warning before.
- Customer has to receive a friendly message before unit expires for reminding to recharge again.

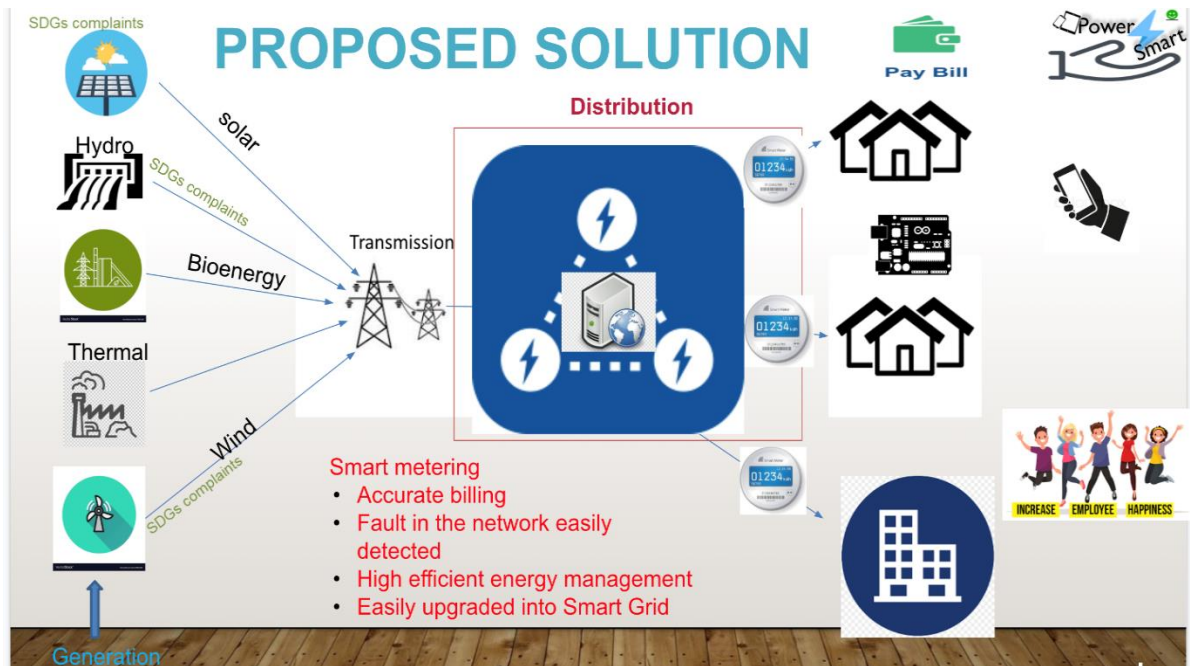


Fig 19: Proposed System

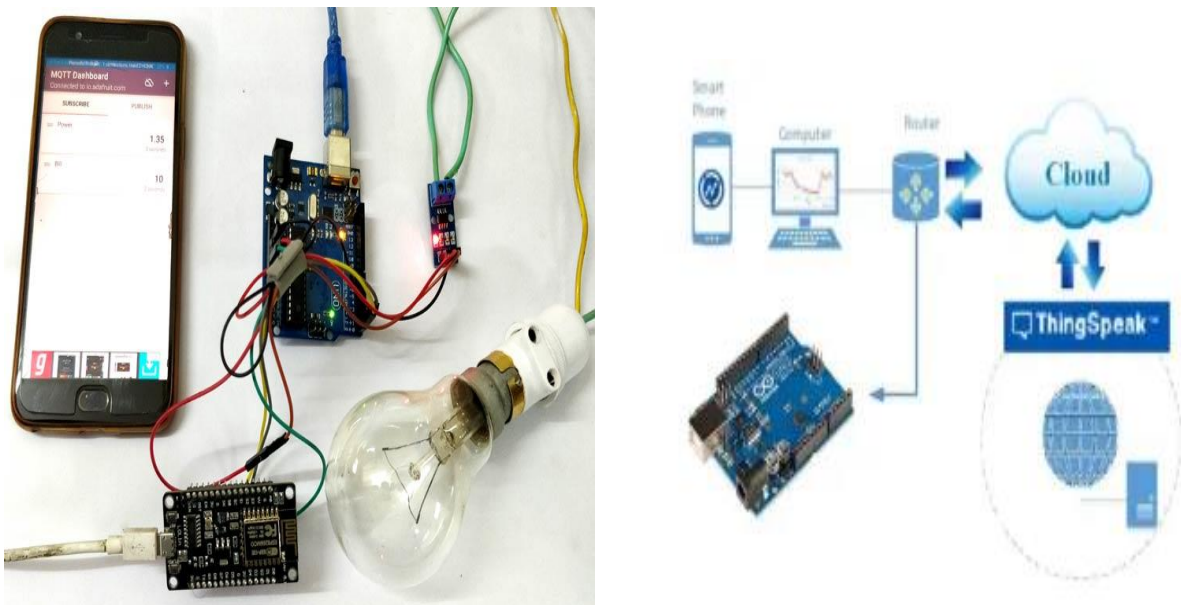


Figure 20: Example of my IoT based prepaid electricity meter devise.

3.5 TANKYU CHART

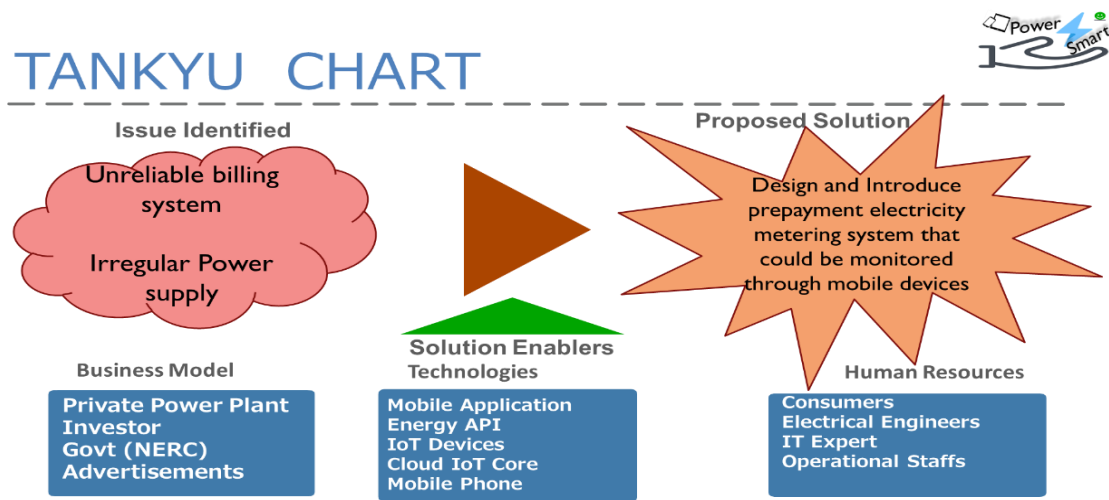


Figure 21: Tankyu Chart

3.6 Research Methodology

This research employed both primary and secondary sources of information. Primary source is through questionnaire design for the study and oral interview of stakeholders in the community. While the secondary sources of information are the relevant textbooks, journals, internet and other research

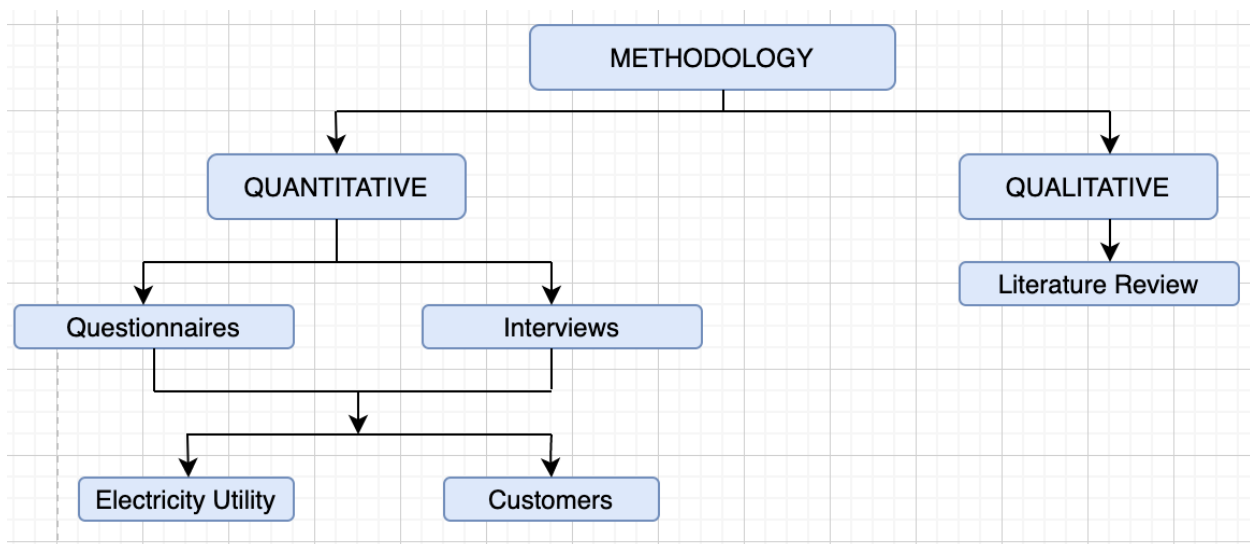


Figure 22: Research Methodology diagram.

Chapter 4

4.1. Implementation, Effects And Validation

This chapter will cover the purpose of designing system, the limitations that were faced during prototyping and the evaluation of the system.

4.2. Design purpose and limitations

Energy shortages is the major challenge facing the industrial sector in Nigeria. Based on network data collected over the period 2011-2015 from three electricity distribution companies (DisCos) drawn from the three major industrial cities of Nigeria. These data were simulated on the Electrical Transient Analysis program (ETAP) Version 12.6. The calculated energy losses for these cities for the said period are 108,959.87 MWH, 149,256 MWH and 72,743.08 MWH respectively. The corresponding revenue losses are N2,434,164,012(USD 6,282,440.164), N3,538,754,758.8(USD 9,132,988.377) and N1,699,751,530.1(USD 4,386,835.0411) respectively.

Feeder	Maximum Energy Loss (MWH) per Feeder					Total Energy Loss (MWH) 2011-2015
	2011	2012	2013	2014	2015	
SabonGari	1,444.0	2,627.9	3,850.4	4,205.6	4,887.2	17,015.1
Bompai	754.53	1,649.4	2,115.9	2,622.3	2,909.1	10,051.2
Independence	2,190.4	3,294.6	3,577.6	3,647.7	4,101.5	16,811.7
Gezewa	2,357.2	4,503.2	4,862.6	4,959.4	5,002.1	21,684.5
Abuja	1,849.5	2,597	3,665.7	3,763.1	4,277.4	16,152.6
Flour Mill	2,765.1	3,889.3	3,997.7	4,087.3	4,242.2	18,981.5
Fagge	1,849.5	2,897	3,665.7	3,763.1	4,477.4	16,652.6
Murtala	1,093.0	2,426.7	2,659.2	3,654.1	3,899.7	13,732.6
Brigade	1,830.1	3,176.5	3,600.4	4,528.7	5,038.5	18,174.2
Total	16,133.33	27,061.6	31,995.2	35,231.3	38,835.1	149,256

Table 8: Maximum Energy Loss In Sabo Gari/Dakata For 2011-2015

Feeder	Maximum Energy Loss (MWH) per Feeder					Total Energy Loss (MWH) 2011-2015
	2011	2012	2013	2014	2015	
Unilever	1,784.15	3,179.31	4,467.64	4,489.87	4,490.49	18,411.46
Evans	998.52	2,128.32	3,208.78	3,527.4	4,002.9	13,865.92
P&G	1,945.42	3,371.58	3,388.19	4,116.05	5,033.58	17,854.82
Beecham	1,636.13	3,143.24	3,232.53	3,358.58	4,207.65	15,578.13
Nestle	1,093.02	2,426.65	2,659.17	3,654.07	3,899.68	13,732.59
OPIC	2,060.2	3,135.84	3,611.95	4,503.56	5,308.46	18,620.01
Guinness	964.71	2,026.78	2,220.05	2,690.31	2,995.09	10,896.94
Total	10,482.15	19,411.72	22,788.31	26,339.84	29,937.85	108,959.87

Table 9: Maximum Energy Loss in Agbara/Badagry for 2011-2015

Energy shortages result from technical (load and no-load losses, line losses etc.) and non-technical losses (losses due to vandalism, illegal connections to transmission/distribution lines or energy theft, meter tampering etc.). In 2014 alone, about 12% of energy losses in the Nigeria distribution network was due to technical losses. The established correlation between electricity supply and Gross Domestic Product makes it necessary, therefore, to ascertain energy losses due to technical losses and the cost implications in the Nigerian distribution network in view of perennial shortages

in electricity supply to the nation's industrial sector. The estimation of the quantity and cost of energy losses in the distribution systems due to vandalism and illegal connections to distribution lines, poor metering where people delayed paying their bills and some of them never pay them, all these too are important factors of electricity losses; hence I took a decision to make this research project so that I can introduce an IoT based prepayment electricity metering system in Nigeria and very sooner as possible. That is the reason why I design this system and make an electricity meter devise that will be used as tool to achieve that purpose of eliminating estimated billings and improve electricity supply to consumers in Nigeria.

In our Lab experiment, I have tested my devise by making a demonstration, the results proved that my sensor (current sensor) sense and control the current flow in a wide variety of applications such as switching mode power supplies, digital watt meters over-current protection circuits etc., the maximum AC or DC that can be detected can reach 5A. this would help me to measure the quantity of electricity consumed by the customer so that he can be charged accordingly.



Fig 23: Experimental hardware arrangement and information display about number of units consumption along with initialization process

4.2.1 RFID:

RFID or Radio Frequency Identification System is a technology-based identification system which helps identifying objects just through the tags attached to them, without requiring any line of sight between the tags and the tag reader. All that is needed is radio communication between the tag and the reader. The number of units that were consumed by customer with predefined payment then it will ask for next payment through smart card. Until the payment completion process the loads will be in off state only. After smart card reading process was done it will check about enough balance availability. If the enough balance will be available, then loads will be on.



Fig:24 LCD display showing about status of card reading.

Experimental Results:

The hardware components of the system were assembled as per the requirements of application. In this paper we are integrating the system for payment using RFID tags, Information about consumption can be done by using SMS which will be sent to registered mobile number about consumption of units and cost of consumption. The consumption details will also be saved in server. Here in our implementation we are using free server that was associated with MATLAB i.e thing speak server. We could able to on or off the loads though web by clicking on the links given through thing speak server or sending message as star one or zero[10].



Fig:25 Experimental hardware arrangement and information display about number of units consumption along with initialization process

4.3. The results of my prototype

4.3.1. First Result

I tried to connect all hardware items and checking if every part works each other properly, with this test as it shown below, you can charge your Electricity meter and see the quantity of Electricity you bought in screen or LCD-Display.

Figure 25: Charge Electricity meter or putting Electricity credits in Electricity meter. See liters on the screen 30

4.3.2. Second Result

This demonstration showed how the Electricity flow sensor counting the Electricity pass through the pipe, so below is the image showed how it is before Electricity pass through the sensor, nothing it is yet counted, The Flow rate was: 0.00 L/M, Total Electricity consumed was: 0.00 L.



Fig:26 Before cut-off point

Now these are the readings in app until cut-off point (Fig.26).



Fig:27 Reading before cut-off point

4.3.3. Third Results

Fig. 22 Readings in app before cut-off point

Next, we see the output of the project when it reached cut-off point. When the cutoff value reaches then the kit will automatically be turned off until we change the cut-off point and it is as in below figure (Fig.28).



Fig:28 After cut-off point

Now these are the readings in app and the after cut-off point (Fig. 29)

SENSOR INFORMATION	
LOAD-1	1
LOAD-2	1
TOTAL UNITS	2
COST	6
CUT OFF UNITS	2
SET	

Fig:29 Reading after cut-off point

The data of the proposed system is uploaded into Thing speak software (Figs.30 and 31)

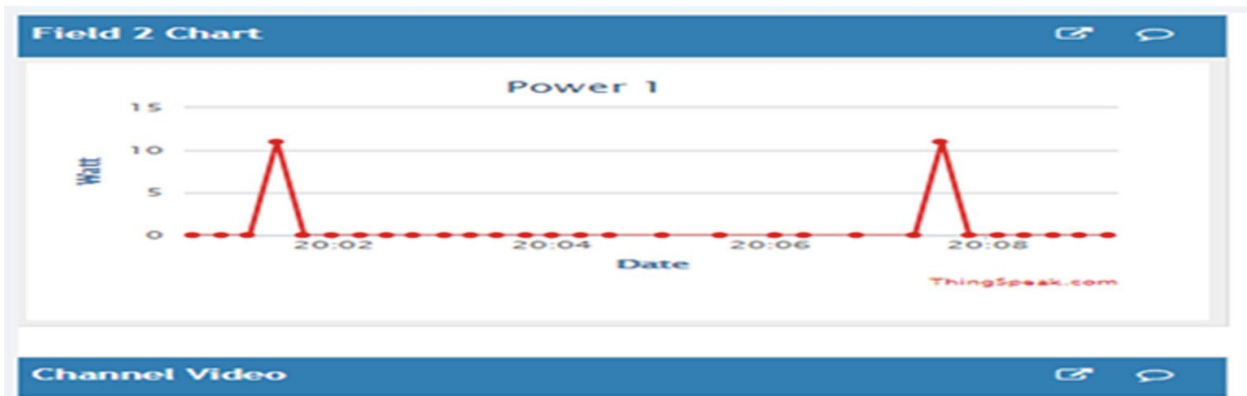


Fig. 30. Power1 vs Time

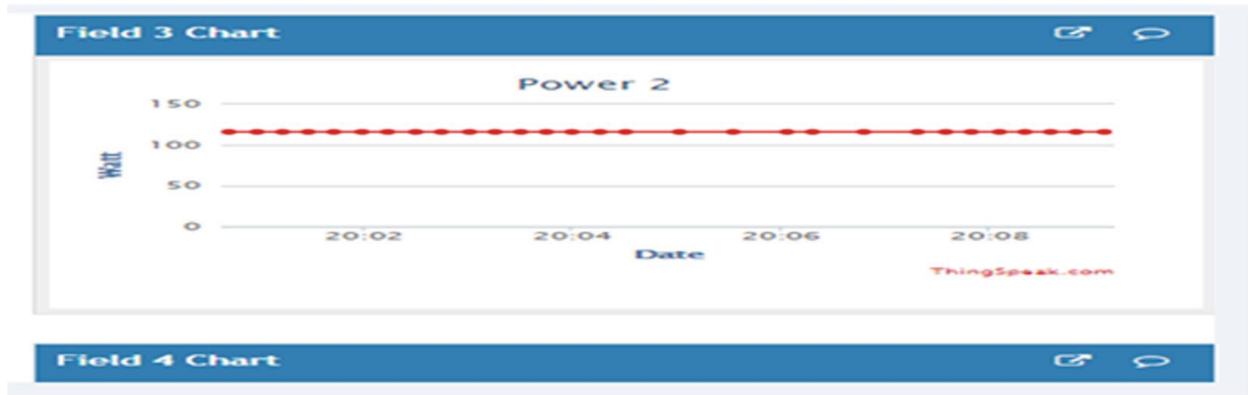


Fig. 31 Power2 vs Time

This demonstration showed how the current sensor is used in measurement of energy consumed, as electricity measurement is the key to business of electricity supply. Measurement is vital to management of electricity value chain because if it cannot be measured it cannot be managed.

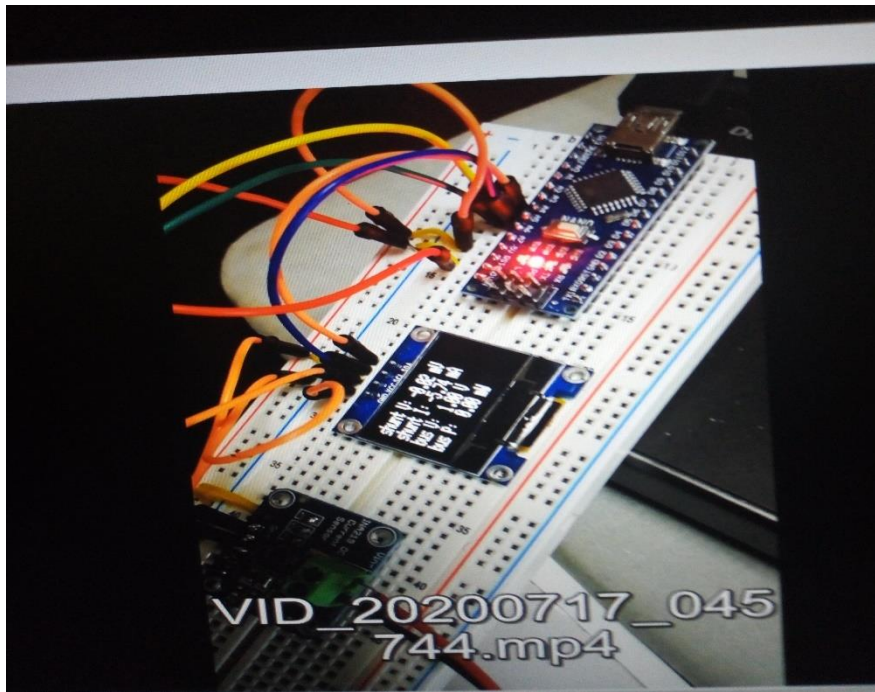


Fig:32 measurement of Energy consumed

This demonstration helped me to evaluate myself and know the prototyping limitation or stage am I now, even though it is still having much to be done but the measuring energy consumption is very crucial part for smart meter prototyping because you cannot know how much money do you charge a client without first knowing how much quantity of energy consumed and then charge accordingly. But the remaining part will be done in future as I kept working on it and improve my prototype.

4.4. The effects of new solution

My System has many effects not only for accurate billing measurement but also prevent fraud of energy theft by introduction of relay to cut-off when the unit loaded is low.

4.4.1. The effects towards energy theft:

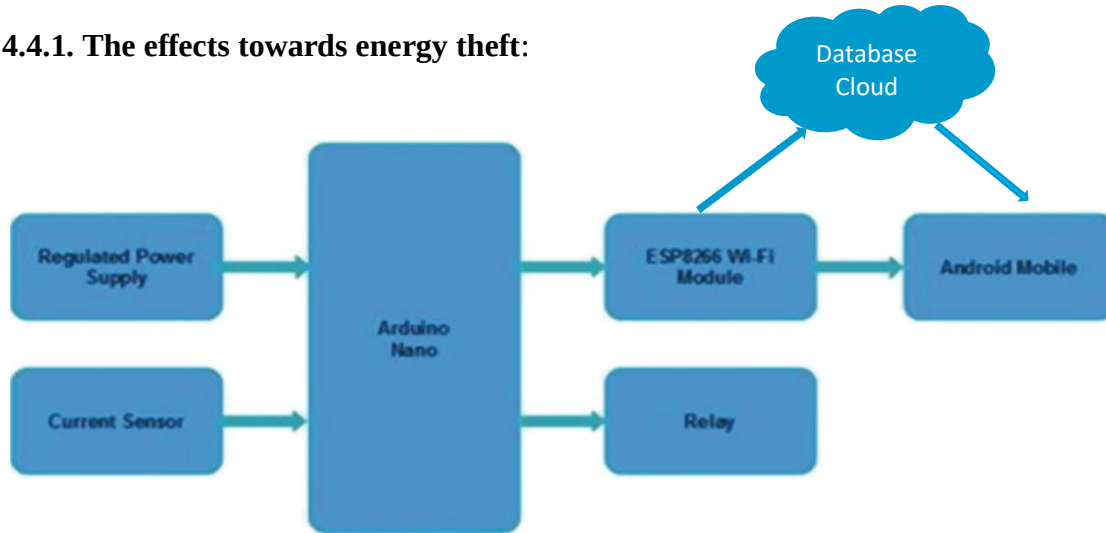


Fig:33 Block Diagram of the proposed system

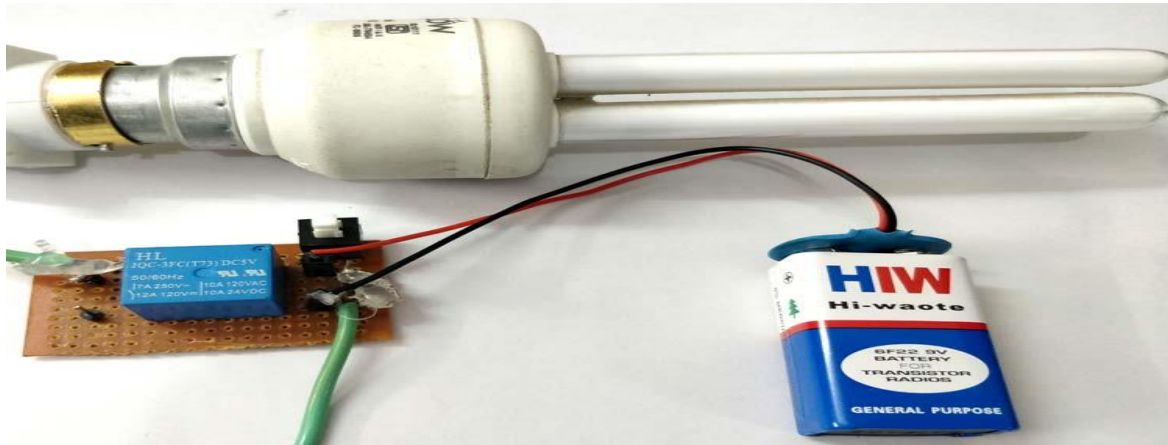


Figure:34 Relay to cut off load when set threshold is reached

The user can get the information about the energy consumption, loads connected, corresponding billed amount through the dedicated app which will be in their smart phones by the end of the month. Then one can directly pay our bill through that app. If bill was not paid then the meter will automatically get turn off. To turn off the meter some threshold value can be set for that. Hence, we can easily control our consumption of electricity and know that how many units that consumer is utilizing on each appliance. This also makes the billing system easier without any errors and intermediate human intervention.

System

- The effect of my solution towards the electricity consumed but not billed.

Consumed by illegal consumers (Pilferage)

Clear event state:

In order to prevent external tamper to meter, meter's valve will be closed automatically when following events happens:

- Passing something to interrupt its operation.
 - Cutting any of the cables.
 - Insufficient battery voltage or removing of battery.
 - Meter restart.
- The effects of my solution towards poor metering (Irregular or Estimated Billing).

My system is a solution for poor metering because it is able to measure domestic, commercial and industrial electricity consumed and shows the used real amount. This is directly link to the database of the energy company to verify with consumers through text messages received on mobile devices. If there is any alteration the system will alert both the registered consumer and the distribution company so that the fault or intruder will be traced. They can act to see what happen in a very early time. And I will keep working on it so that to increase its accuracy in future.

- The effects of my solution towards the Energy consumed and billed
 - ✓ Electricity Billed, but never paid (the bill is ignored) and Paid but delayed

My solution is prepayment electricity metering system hence a customer must pay before the consumption therefore this issue of unpaid bills and delaying paying the bills won't happen again.

4.5. Evaluation of the System and Analysis

After making the first prototype of my system, I made an evaluation for my system by interviewing and sending the questionnaire to be filled out by Electricity utility in Nigeria and its customers in order to get their feedback or to know what they think about my solution so that I can correct or improve my technology solution. Below is the feedback from one of electricity Distribution company in Nigeria and its customers.

4.5.1. Interview and Answers

Research Questions

The following are some of the research Questions developed to obtain data for this study:

Table 10: Evaluation interview and Answers

No	Questions	Concern	Answer
1	Where is your location?	Urban, Sub-urban or Rural	More percentage of people live in Sub-urban
2	What type of Meter did you have?	Billing	More than 50% without electricity meter. Smart will be provided for consumers
3	How many hours of electricity supply did you receive per day?	Power supply regularity	Majority receives 4-8 hrs/day. More supply by implementing this solution since energy losses will be eliminated.
4	What is your billing method?	Billing	More than 50% of electricity consumers are without meter. My solution provides smart meter to consumers without meter and make existing energy meter smart
5	What is the size of your apartment?	Consumption rate	More people live in 3Bed-room apartment
6	What is the average bill payment per month?	Balancing supply with payment	People pays more compared to supply – estimated bills. In my solution, you pay the exact consumption.
7	What is the response rate of billing complaints?	Faults clearance	Unfair but the implementation of smart billing, faults will be easily detected, and response is faster
8	Are you satisfied with the service provider?	Billing	Close to 70% of consumers responded No!. Because they were paying for they do not consumed
9	Do you use a mobile phone (GSM)?	Easy payment and monitoring	100% response. The technology that new solution rides on

10	Did you have regular access to online services e.g Facebook, Google, etc?	Easy payment and monitoring	Close to 90% of mobile subscribers have access
11	Would you prefer a service that allows you to monitor the electricity consumed and billing through your mobile device? (e.g. phone)	Acceptance of the new solution	Over 90% of subscribers prefers the new solution
12	How much are you are willing to pay for this new service?	Service payment	About 70% of people response to the questionnaire ready to pay according to the consumption
13	Would you like to know your weekly/monthly energy consumed via your mobile device?	Eliminating Estimated billing	Consumers prefers to monitor energy usage through their mobile devices. The system makes provision for that.
14	Would you like to buy your electricity pre-paid card via your mobile phone?	Estimated billings	Though the users can do that with existing system, this will be integrated into the solution
15	Did you sometimes notice distortion in the line and irregular meter reading?	Energy theft	The distribution faces problems of energy theft power losses from generation and distribution in the network. This solve through introduction of the new system.
16.	Would you prefer a service that allows you to remotely monitor the Grid meters?	Reducing Power loss	The distribution companies prefers remote monitoring of all infrastructure which new solution can provide by extension
17.	There is a huge loss (transmission, technical, commercial, collection etc) from Energy Generation to	Irregular power Supply	Smart grid solution advice.

	Distribution, will you prefer a system that automatically checkmate these losses and secure the network?		
--	--	--	--

I.SWITCHING DEVICE:

In our system we are using SSR as switching device even though we can use RELAY because SSR is highly advantageous. We are using switching device to switch the energy meter. For ON and OFF purpose of meter we are using switching block. SSR stands for SOLID STATE RELAY. Why SSR instead of RELAY? • Both are used as AC switching device, but if switching speed is high then SSR is suitable, if switching speed is slow then RELAY is used. • Relay life decreases as number of usage time increases, but in SSR there is no change. • For driving RELAY, current or power required is more comparatively to SSR. • For switching SSR requires 15amp, whereas RELAY needs (30amp,50amp,90amp) as per requirement.

IV.EQUATIONS AND MATHEMATICAL CALCULATION

Our system does not contain very vast and difficult calculations. Usually different meters have different readings. Some have,

$$1,500 \text{ blinks} = 1 \text{ unit}$$

Mostly, 3200 blinks = 1 unit depends on manufacturer.

In our case 3200 blinks of LED is 1 unit. But for practical purpose,

Assumption we made in our system,

$$5 \text{ blinks} = 1 \text{ unit of power consumption.}$$

Let, X = number of blinks of LED

Y = number of units of electricity.

Z = cost of consumption.

Basically,

$$\text{No. of units (Y)} = (X/3200)$$

But in our case,

$$Y = (X/5)$$

Assume that 1 unit cost = 5rs.

$$Z = Y * 5\text{rs}$$

For Threshold,

Assumed threshold set value will be = 5 units for practical.

If units reach,

Threshold value – 1 unit = 5-1 = 4 units,

Notification will be sent to consumer, if consumer doesn't react and increase the threshold value then meter will automatically get OFF. Again, to turn it ON consumer has to visit

webpage again to increase threshold value. For practical purpose increment and decrement of threshold can be done by +5units or -5units.

Normally, basic unit of electricity is Kilowatt hour (KWh).

1kWh = 1000 watt for 1 hour.

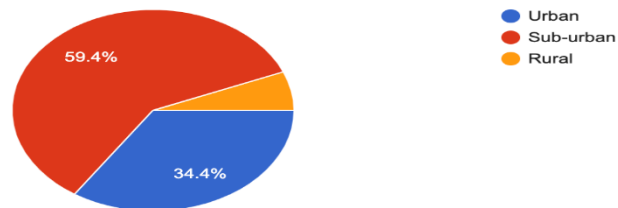
Example, Ten 100watt bulbs used for 1 hour gives 1kWh.

4.5.2. Questionnaire and Answers

Below are the questions and responses statistics feedback from Electricity Distribution company and customers that we generated by using google form.

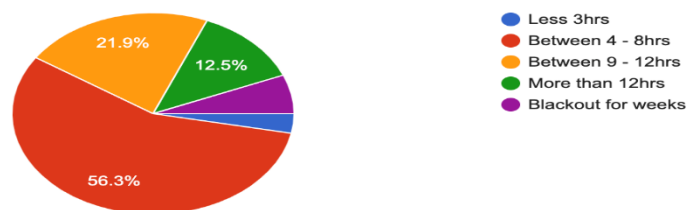
3. Location

32 responses



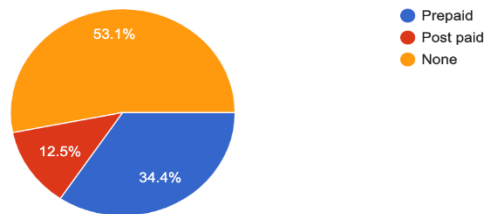
4. Range of hours of supply per day

32 responses



5. What type of Meter did you have?

32 responses



6. What is your billing method

32 responses

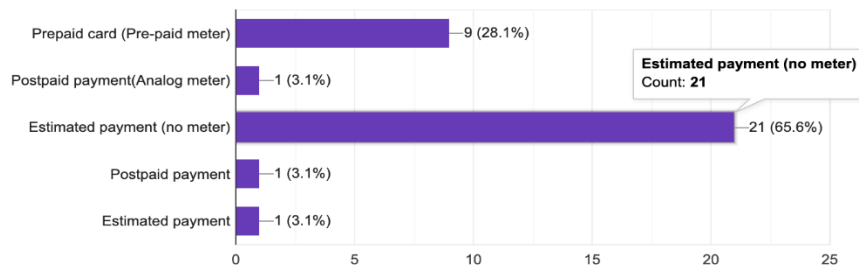
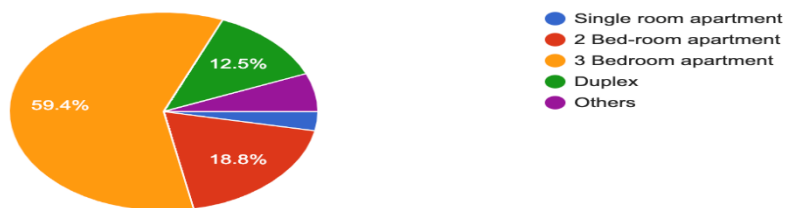


Fig 35: Evaluation level of importance of the system for consumers

7. Size of apartment

32 responses



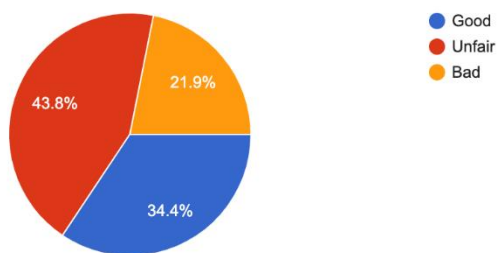
8. What is the average bill payment per month?

32 responses



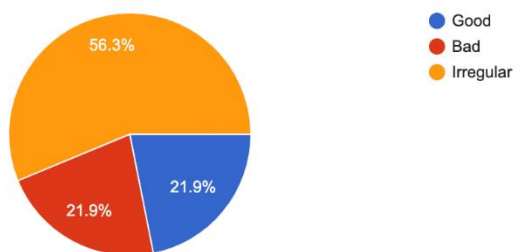
9. The response rate of billing complaints

32 responses



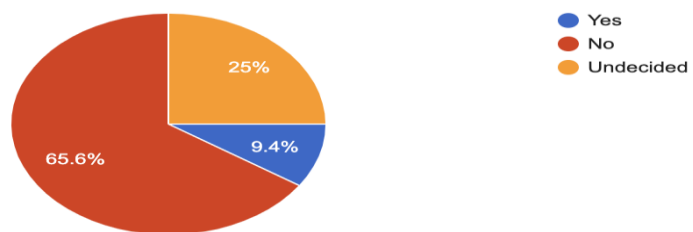
10. The response rate of fault clearing

32 responses



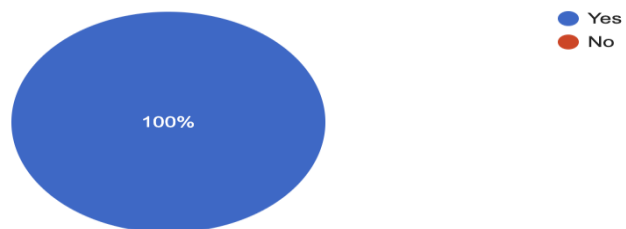
11. Are you satisfied with the service provider?

32 responses



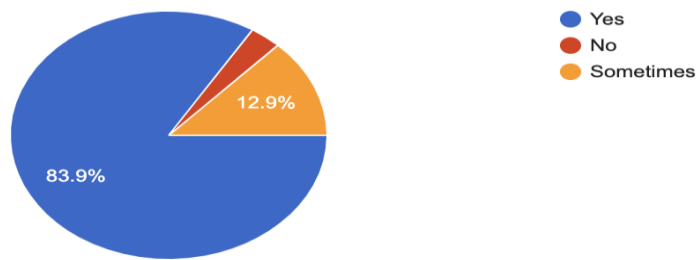
12. Do you use a mobile phone (GSM)?

31 responses



13. Did you have regular access to online services e.g Facebook, Google, etc?

31 responses



14. Would you prefer a service that allows you to monitor the electricity consumed and billing through your mobile device? (e.g. phone)

32 responses

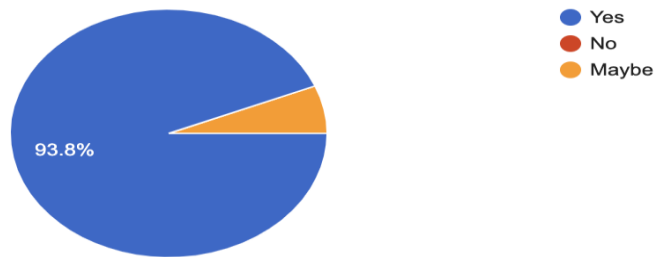
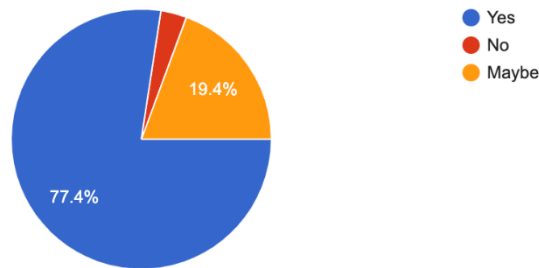


Fig 36: Evaluation level of importance of the system for the electricity distribution company

17. Would you like to buy your electricity pre-paid card via your mobile phone?

31 responses



Any Comment? For further correspondence, please send a message to smrtbillings@gmail.com

14 responses

Wondering why prepaid meter is unavailable, and difficult to get.

Is this a new innovation?

Energy supply has not been constant. I would love to see improvement in this regard.

Regular power supply to the societies will usher in tremendous growth of businesses eg the SME and also contribute to the well beings of the citizens. International investors will develop interest in the country which will lead to increase in the GDP of the country at the long run. Crime rates will decrease since many youths and able men and women will be fully engaged in different vocations. There will not be any hidden places for criminals to operate as all dark spots will be illuminated and our security operatives' jobs will be less tasking. Thanks.

4.5.3. Analysis

The feedback statistics above shows that 93.8% agreed that my new system is very important for customers to monitor the electricity consumed and billings through mobile device and will highly eliminate estimated billing and allow the distribution company to monitor consumers' activities and also to prevent energy theft. The same, 93.8% are willing to pay for the new system because it is far cheaper than present solution of estimated billing and also, they spend more on running generators to make up for irregular supply. Though initial cost procuring a single – 3 phase smart meter is (USD10 – 20) if we are purchasing large quantities. So, it cost much less than USD 20 to procure the hardware to upgrade the existing energy meter to smart.

In brief, my new system is very needed and seems to make an impact to both electricity distribution companies and customers, but we still need to improve it based on the feedback and needs from Distribution companies and consumers.

4.5.4 Limitations

The following are some of the limitation identified

- ◆ Existing energy companies (DisCos) might feel threaten but the mitigation is to make them part of the stakeholders
- ◆ Installation of Wi-Fi/GPRS enabled meters at consumer end will not be possible over a certain period. It takes a lot time to do. Manpower requirement will be high at the time of installation and initialization. The cost of implementation will be high.
- ◆ The requirements of the system vary based on place the consumer need the smart energy meter. According to the requirements given by consumer need to develop the hardware and it should be configured according to the rules and regulations framed by that state government
- ◆ As the unit charges will be varies according to the category of supply and power distribution units.
- ◆ This system will have many drawbacks in hill stations, forests and the places where the network coverage problems arise.

4.5.5 Plan and Future Work

Most of the off-grid electrification project in Nigeria have failed due to lack of technical operations and maintenance tariff collections mechanism.

This project will be upgraded to smart grid management systems and extend to other Power Generating plants, so that consumers can have overall accurate billing system in the country through the applications of ICT.



Chapter 5

5.1 Conclusion and Recommendation

This chapter includes a conclusion of the important to what I have done, why I made it and what I am planning in the future. IoT based prepayment smart billing was born out of the necessity to provide lasting energy management solutions to the irregular power supply and billings management challenges in Nigeria. My aim is to develop a technologically advanced electricity meter reading systems for distribution companies in Nigeria to leverage on the power of technology in order to provide end-to-end solutions for energy management, thus enabling citizens to have uninterrupted power supply and smooth billings. This will also enable end users to understand their usage patterns and adjust to suite their budget. by closely monitoring electricity consumption, both DisCo and consumers alike can establish bi-directional communication and enable concerted efforts in managing this precious resource by taking full advantage of the existing technologies at the disposal of both the utilities and consumers. I have already started making an IoT based prepayment electricity metering system that will help to achieve this goal, I finished to make a circuit design, hardware assembling and console, a current sensor programming in order to know how much amount of energy that is being consumed by customer in order to charge for money accordingly to the consumed quantity of electricity and now am working again on programming to console with Wi-Fi component and relay switch so that it can switch off electricity when credit expires or switch on when it recharged again, in future. I will work on IoT part so that I can send a friendly remind message to tell a customer that his credit is become low so that he can recharge meter before it will be switched off.

After conducting a survey and established several factors affecting both the utility company and the consumer, I recommend both electricity Distribution Companies (DisCos) and customers to use this kind technology written in my thesis in order to overcome some challenges such as:

1. In some cases, consumers are not well informed of the energy measurements used by the distribution companies and how they can contribute to safeguarding against energy loss
2. The distribution energy provider can only plan in retrospect as there is a need to wait for all bill information to be collected for any decision making to be done
3. There is only a one-way communication between the distribution companies and the consumer hence the distribution companies will only sometimes receive feedback in the form of a bill conflict or when there is a service shutdown. It would be ideal to have real-time feedback between the customer and the operator to minimize losses and bill shock on the part of the consumer.

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APPENDICES

APPENDIX A

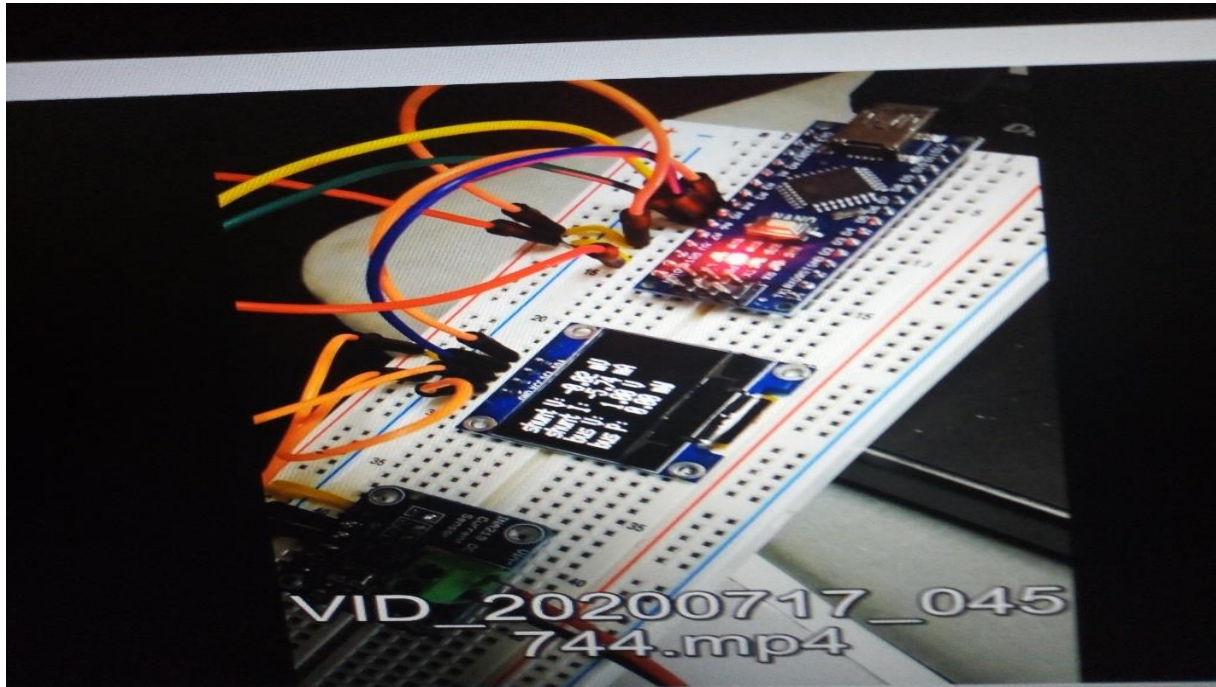


Figure A: Current sensor and total consumed energy measurement Results.

APPENDIX B

```
Arduino File Edit Sketch Tools Help
water_flow_sensors_new_1 | Arduino 1.8.8

water_flow_sensors_new_1
{
  X = pulseIn(input, HIGH);
  Y = pulseIn(input, LOW);
  TIME = X + Y;
  FREQUENCY = 1000000/TIME;
  WATER = FREQUENCY/7.5;
  LS = WATER/60;
  if(FREQUENCY == 0)
  {
    if(!isnan(FREQUENCY))
    {
      lcd.clear();
      lcd.setCursor(0,0);
      lcd.print("VOL. :0.00");
      lcd.setCursor(0,1);
      lcd.print("TOTAL:");
      lcd.print(TOTAL);
      lcd.print(" L");
    }
    else
    {
      TOTAL = TOTAL + LS;
      Serial.print(FREQUENCY);
      lcd.clear();
      lcd.setCursor(0,0);
      lcd.print("VOL.: ");
      lcd.print(WATER);
      lcd.print(" L/h");
      lcd.setCursor(0,1);
      lcd.print("TOTAL:");
      lcd.print(TOTAL);
      lcd.print(" L");
    }
  }
  delay(1000);
}
```

Figure B: The source code of current measurement and total consumed energy measurement.
APPENDIX C

```

Arduino File Edit Sketch Tools Help
water_flow_sensors_new_1 | Arduino 1.8.8

water_flow_sensors_new_1
#include <LiquidCrystal.h>
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);
int X;
int Y;
float TIME = 0;
float FREQUENCY = 0;
float WATER = 0;
float TOTAL = 0;
float LS = 0;
const int input = A0;
void setup()
{
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Water Flow Meter");
  lcd.setCursor(0, 1);
  lcd.print("*****");
  delay(2000);
  pinMode(input, INPUT);
}
void loop()
{
  X = pulseIn(input, HIGH);
  Y = pulseIn(input, LOW);
  TIME = X + Y;
  FREQUENCY = 1000000/TIME;
  WATER = FREQUENCY/7.5;
  LS = WATER/60;
  if(FREQUENCY >= 0)
  {
    if(!isnan(FREQUENCY))
    {
      lcd.clear();
      lcd.setCursor(0, 0);
    }
  }
}

```

Figure C: The next of source code of electricity and total consumed energy measurement.

APPENDIX D : Source code of login interface to accessing energy data

Web Interface login page index.php

```

<html>

<head>

<!-- Global site tag (gtag.js) - Google Analytics -->
<script async src="https://www.googletagmanager.com/gtag/js?id=UA-174505637-1"></script>
<script>
  window.dataLayer = window.dataLayer || [];
  function gtag(){dataLayer.push(arguments);}
  gtag('js', new Date());
  gtag('config', 'UA-174505637-1');
</script>
<title> Display Energy Data</title>
</head>

```

```

<body bgcolor='lightcyan'>
<br>
<h2 align="center"> <font color="red">Smart Billing system data</font> </h2><br></head>
<center></center><br>
<h2 align="center"> Login </h2>
<br>
<center>
  <form action="login.php" method="post">
    <div>
      <label for="username">Username:</label>
      <input type="text" id="username" name="username">
    </div>
    <br>
    <div>
      <label for="pass">Password :</label>
      <input type="password" id="pass" name="password"
        required>
    </div>
    <br>
    <input type="submit" value="Sign in">
  </form>
</center>
</body>
</html>

```

Smart Billing system data



Login

Username:

Password :

Figure D: Login page to accessing energy data

Display data page IndexLogin. php

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<script language="javascript">
```

```
    setTimeout(function() {
```

```
        window.location.reload();
```

```
    }, 60000);
```

```
</script>
```

```
<!-- Global site tag (gtag.js) - Google Analytics -->
```

```
<script async src="https://www.googletagmanager.com/gtag/js?id=UA-174505637-1"></script>
```

```
<script>
```

```
    window.dataLayer = window.dataLayer || [];
```

```
    function gtag(){dataLayer.push(arguments);}
```

```

</script>

<title> Display Energy Data</title>

</head>

<body bgcolor='lightcyan'>

<h4 align="right">Welcome <?php echo $_SESSION["user"]; ?></h4>

<h2 align="center"> <font color="red">Smart Billing system data</font> </h2><br></head>

</h2><br>

<center></center><br>

<h2 align="center"> Data collected by the Energy Sensor </h2>

<br>

<h3><!--a href="display-linegraph.html" target="_blank">Line Graphs</a> <a href="display-bargraph.html"
target="_blank">Bar Graphs</a--></h3>

<?php
/*
Abraham Olufemi
*/

$servername = "fdb18.awardspace.net";
// REPLACE with your Database name
$dbname = "3348988_temphumidity";
// REPLACE with Database user
$username = "3348988_temphumidity";
// REPLACE with Database user password
$password = "living2018";
// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT id, User, location, dateTime, voltage, current, power,energy FROM electricityData ORDER BY id
DESC";

echo '<table cellpadding="5" cellspacing="5" border="1" align="center">

<tr>

```

```

<td>ID</td>
<td>User</td>
<td>Location</td>
<td>Reading Time</td>
<td>Voltage (V)</td>
<td>Current(mA)</td>
<td>Power (mWatt)</td>
<td>Energy (mwatth)</td>
</tr>';

//data to collect json
$data = array();
if ($result = $conn->query($sql)) {
    while ($row = $result->fetch_assoc()) {
        $row_id = $row["id"];
        $row_user = $row["User"];
        $row_location = $row["location"];
        $row_dateTime = $row["dateTime"];
        $row_voltage = $row["voltage"];
        $row_current = $row["current"];
        $row_power = $row["power"];
        $row_energy = $row["energy"];
        //$row_reading_time = $row["reading_time"];
        // Uncomment to set timezone to - 1 hour (you can change 1 to any number)
        //$row_reading_time = date("Y-m-d H:i:s", strtotime("$row_reading_time - 1 hours"));

        // Uncomment to set timezone to + 4 hours (you can change 4 to any number)
        //$row_reading_time = date("Y-m-d H:i:s", strtotime("$row_reading_time + 4 hours"));
        echo '<tr>
            <td>' . $row_id . '</td>
            <td>' . $row_user . '</td>
            <td>' . $row_location . '</td>
            <td>' . $row_dateTime . '</td>
            <td>' . $row_voltage . '</td>

```

```
<td>' . $row_current . '</td>
<td>' . $row_power . '</td>
<td>' . $row_energy . '</td>
</tr>';
}
$result->free();
}
$conn->close();
echo'</table>';
    //print json_encode($data);
?>
</body>
</html>
```

Connexion Fails, Check your username and password and Try again

Figure E: Login page with wrong username/password

Smart Billing system data



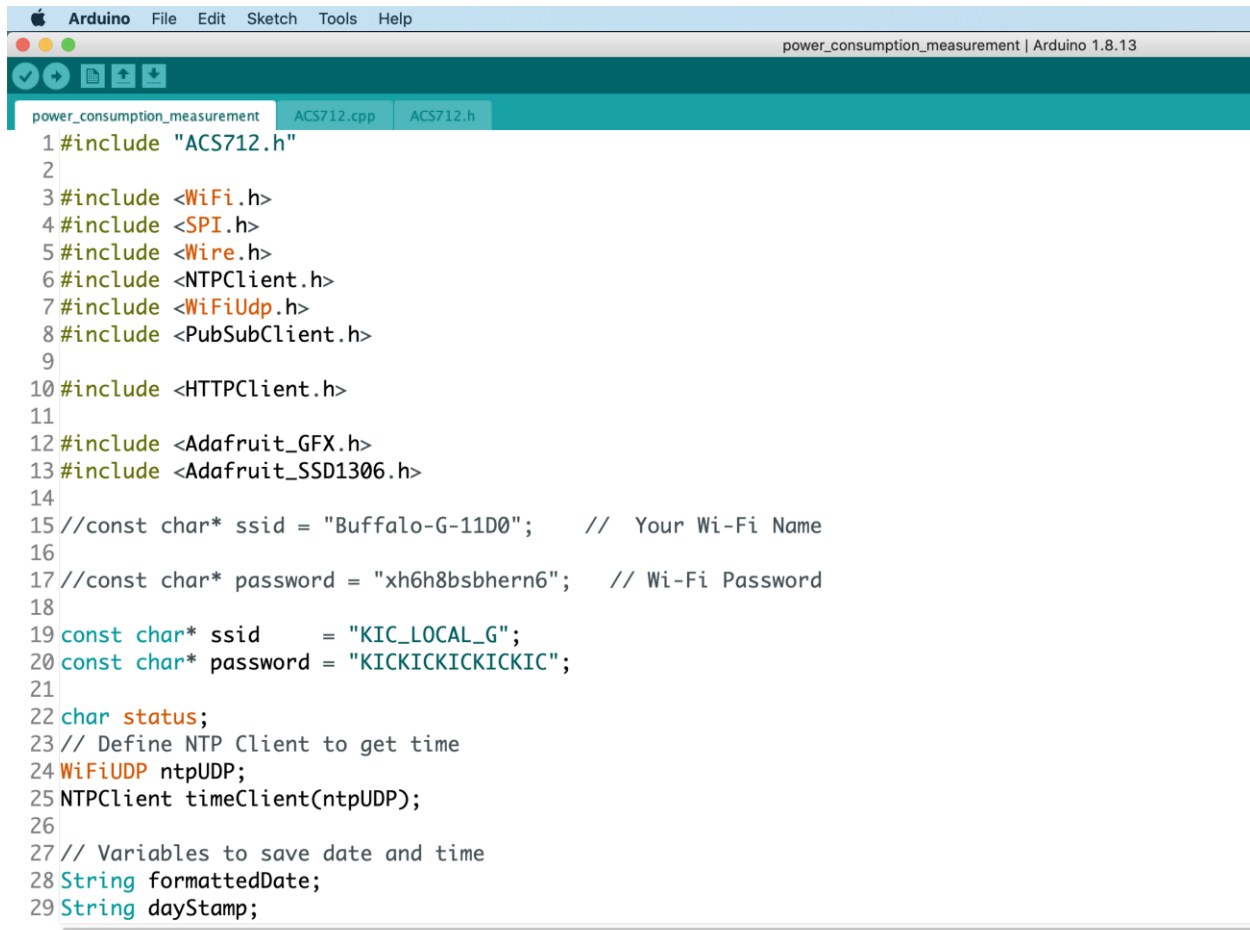
Data collected by the Energy Sensor

ID	User	Location	Reading Time	Voltage (V)	Current(mA)	Power (mWatt)	Energy (mwatt)
203	Abraham	JAPAN	2020-08-06 17:44:43	220	2.72	598.83	776.02
202	Abraham	JAPAN	2020-08-06 17:43:41	220	2.68	588.99	765.66
201	Abraham	JAPAN	2020-08-06 17:42:39	220	2.82	620.81	755.51
200	Abraham	JAPAN	2020-08-06 17:41:37	220	2.85	627.19	744.81
199	Abraham	JAPAN	2020-08-06 17:40:35	220	2.8	615.87	733.96
198	Abraham	JAPAN	2020-08-06 17:39:33	220	3.09	678.81	723.36
197	Abraham	JAPAN	2020-08-06 17:38:30	220	2.63	578.77	711.56
196	Abraham	JAPAN	2020-08-06 17:37:28	220	2.64	581.74	701.57
195	Abraham	JAPAN	2020-08-06 17:36:26	220	2.59	569.05	691.53

Figure E: Login page to accessing energy data

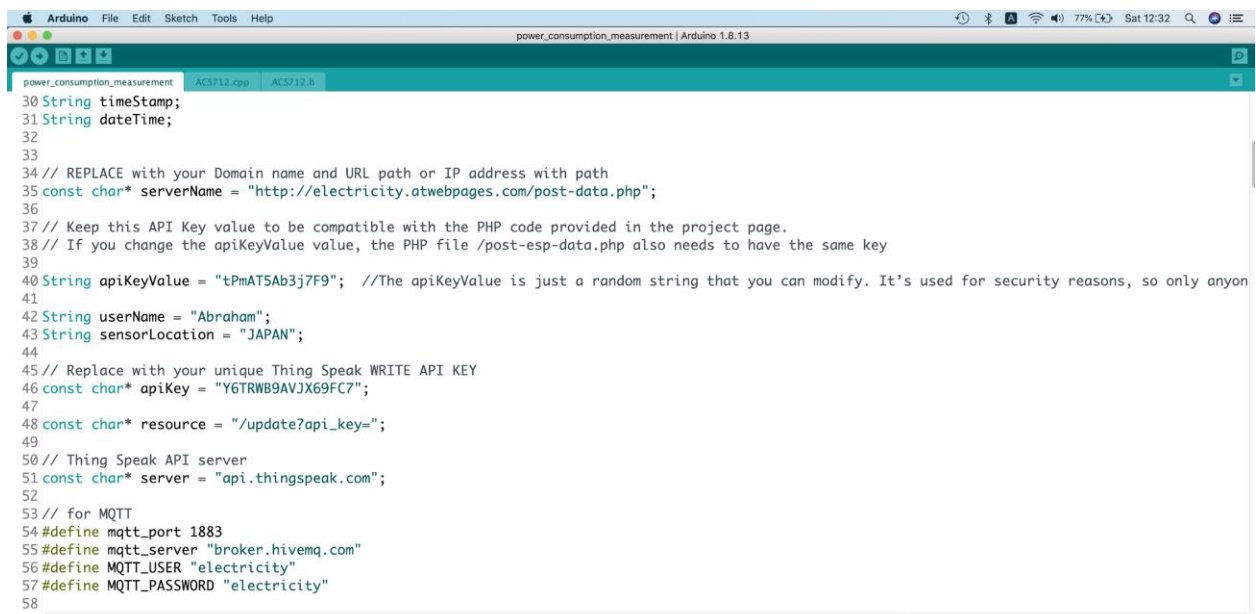
194	Abraham	JAPAN	2020-08-06 17:35:24	220	2.68	589.44	681.69
193	Abraham	JAPAN	2020-08-06 17:34:22	220	2.83	622.09	671.54
192	Abraham	JAPAN	2020-08-06 17:33:20	220	2.82	621.23	660.78
191	Abraham	JAPAN	2020-08-06 17:32:15	220	2.68	590.34	649.72
190	Abraham	JAPAN	2020-08-06 17:31:02	220	2.83	622.51	637.6
189	Abraham	JAPAN	2020-08-06 17:29:59	220	2.8	615.87	626.87
188	Abraham	JAPAN	2020-08-06 17:28:57	220	2.76	606.97	616.21
187	Abraham	JAPAN	2020-08-06 17:27:55	220	2.78	612.41	605.65
186	Abraham	JAPAN	2020-08-06 17:26:52	220	2.84	625.7	595.07
185	Abraham	JAPAN	2020-08-06 17:25:50	220	2.77	608.94	584.23
184	Abraham	JAPAN	2020-08-06 17:24:48	220	2.88	632.66	573.74
183	Abraham	JAPAN	2020-08-06 17:23:46	220	2.92	641.62	562.81
182	Abraham	JAPAN	2020-08-06 17:22:43	220	2.86	629.51	551.7
181	Abraham	JAPAN	2020-08-06 17:21:41	220	2.99	658.57	540.87
180	Abraham	JAPAN	2020-08-06 17:20:39	220	3.29	724.58	529.49
179	Abraham	JAPAN	2020-08-06 17:19:35	220	3.12	686.39	516.59
178	Abraham	JAPAN	2020-08-06 17:18:33	220	3.19	701.12	504.66
177	Abraham	JAPAN	2020-08-06 17:17:30	220	3.02	664.38	492.52
176	Abraham	JAPAN	2020-08-06 17:16:28	220	3.17	696.37	481.08
175	Abraham	JAPAN	2020-08-06 17:15:26	220	3.12	686.97	469.06
174	Abraham	JAPAN	2020-08-06 17:14:24	220	3.19	702.45	457.16

Figure F: Sensor data uploaded from the database after successful login.



```
1 #include "ACS712.h"
2
3 #include <WiFi.h>
4 #include <SPI.h>
5 #include <Wire.h>
6 #include <NTPClient.h>
7 #include <WiFiUDP.h>
8 #include <PubSubClient.h>
9
10 #include <HTTPClient.h>
11
12 #include <Adafruit_GFX.h>
13 #include <Adafruit_SSD1306.h>
14
15 //const char* ssid = "Buffalo-G-11D0";    // Your Wi-Fi Name
16
17 //const char* password = "xh6h8bsbhern6";    // Wi-Fi Password
18
19 const char* ssid    = "KIC_LOCAL_G";
20 const char* password = "KICKICKICKICKIC";
21
22 char status;
23 // Define NTP Client to get time
24 WiFiUDP ntpUDP;
25 NTPClient timeClient(ntpUDP);
26
27 // Variables to save date and time
28 String formattedDate;
29 String dayStamp;
```

Fig G: Arduino Code screenshots



```
30 String timeStamp;
31 String dateTime;
32
33
34 // REPLACE with your Domain name and URL path or IP address with path
35 const char* serverName = "http://electricity.atwebpages.com/post-data.php";
36
37 // Keep this API Key value to be compatible with the PHP code provided in the project page.
38 // If you change the apiKeyValue value, the PHP file /post-esp-data.php also needs to have the same key
39
40 String apiKeyValue = "tPmAT5Ab3j7F9"; //The apiKeyValue is just a random string that you can modify. It's used for security reasons, so only anyone
41
42 String userName = "Abraham";
43 String sensorLocation = "JAPAN";
44
45 // Replace with your unique Thing Speak WRITE API KEY
46 const char* apiKey = "Y6TRWB9AVJX69FC7";
47
48 const char* resource = "/update?api_key=";
49
50 // Thing Speak API server
51 const char* server = "api.thingspeak.com";
52
53 // for MQTT
54 #define mqtt_port 1883
55 #define mqtt_server "broker.hivemq.com"
56 #define MQTT_USER "electricity"
57 #define MQTT_PASSWORD "electricity"
58
```

```

Arduino File Edit Sketch Tools Help
power_consumption_measurement | Arduino 1.8.13

power_consumption_measurement ACS712.cpp ACS712.h
59 //for the current sensor
60 char watt[5];
61 ACS712 sensor(ACS712_30A, 33);
62 long current_time = 0;
63 float Wh=0, I=0;
64 float V= 220; //We use 230V because it is the common standard in European countries, Change to your local, if necessary
65 float P; // variable to store the power
66
67 #define SCREEN_WIDTH 128 // OLED display width, in pixels
68 #define SCREEN_HEIGHT 64 // OLED display height, in pixels
69
70 // Declaration for an SSD1306 display connected to I2C (SDA, SCL pins)
71 #define OLED_RESET -1 // Reset pin # (or -1 if sharing Arduino reset pin)
72 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);
73
74 //setup function begins here
75 void setup() {
76
77 // put your setup code here, to run once:
78 Serial.begin(115200);
79 // initialize with the I2C addr 0x3C
80 if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)){
81 Serial.println("SSD1306 allocation failed");
82 for(;;);
83 }
84
85 // Clear the buffer.
86 display.clearDisplay();
87 sensor.calibrate();
88
89 //connect to Wifi
90
91 setup_wifi();
92
93 // Initialize a NTPClient to get time
94 timeClient.begin();
95 // Set offset time in seconds to adjust for your timezone, for example:
96 // GMT +1 = 3600
97 // GMT +8 = 28800
98 // GMT -1 = -3600
99 // GMT 0 = 0
100 timeClient.setTimeOffset(32400);
101
102 // Display Text
103 display.setTextSize(1);
104 display.setTextColor(WHITE);
105 display.setCursor(0,28);
106 display.println("Smart Billing");
107 display.println("");
108 display.println("Abraham");
109 display.display();
110 }
111
112 void loop() {
113 // put your main code here, to run repeatedly:
114
115 I = sensor.getCurrentAC();
116 P = V * I; // compute power

```

Figure H: Login page to accessing energy data

```

Arduino File Edit Sketch Tools Help
power_consumption_measurement | Arduino 1.8.13
power_consumption_measurement ACS712.cpp ACS712.h
117 long last_time = current_time;
118 current_time = millis();
119 Wh = Wh+ P *(( current_time -last_time) /3600000.0) ; // convert power into whatt h
120 dtostrf(Wh, 4, 2, watt); //Now, we have to convert this Wh into character form to send it to NodeMCU,
121 //dtostrf(floatvar, StringLengthIncDecimalPoint, numVarsAfterDecimal, charbuf);
122 Serial.print("Current = ");
123 Serial.print(I);
124 Serial.println(" A");
125 Serial.print("Energy = ");
126 Serial.print(watt);
127 Serial.println(" watth");
128
129 // display on LCD
130 display.clearDisplay();
131 display.setCursor(0,0);
132 display.setTextSize(1);
133 display.println("Smart Billing");
134 display.println("");
135 display.println("Abraham");
136 display.println("");
137 display.print("I:");
138 display.print(I);
139 display.print(" mA");
140 display.print(", P:");
141 display.print(P);
142 display.println(" mWatt");
143 display.println("");
144 display.print("E:");
145 display.print(watt);

```

Figure I: Code for ESP32 on IDE to read current data

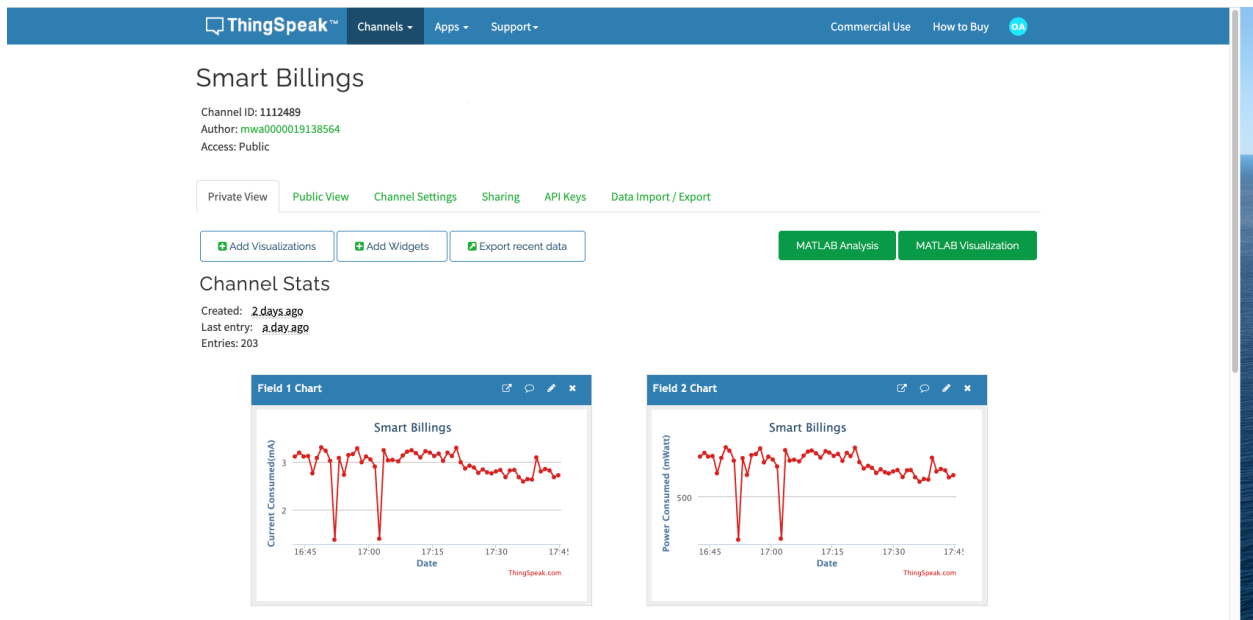


Figure J: Thingspeak visualizations screenshots

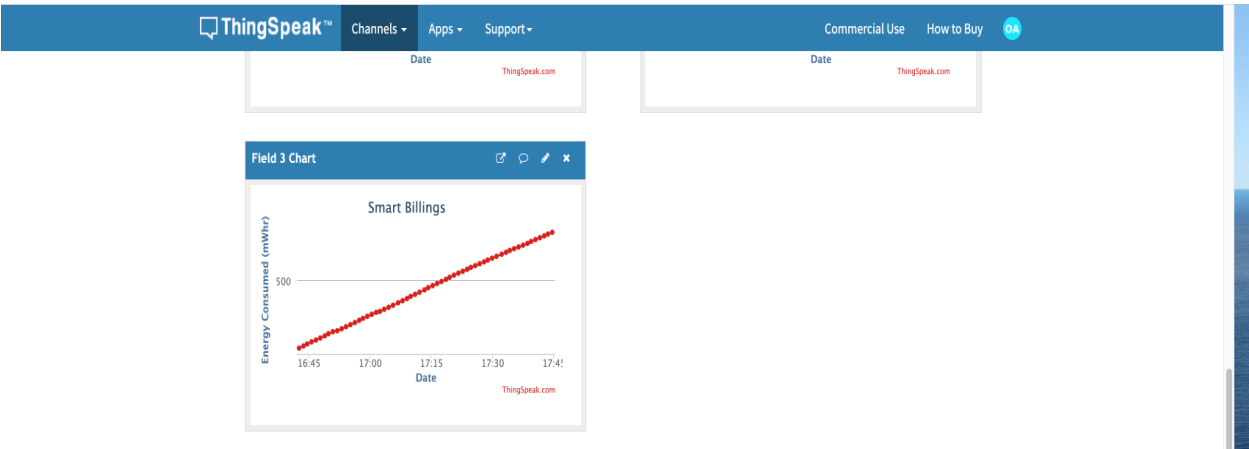


Figure K: Thingspeak visualizations screenshots

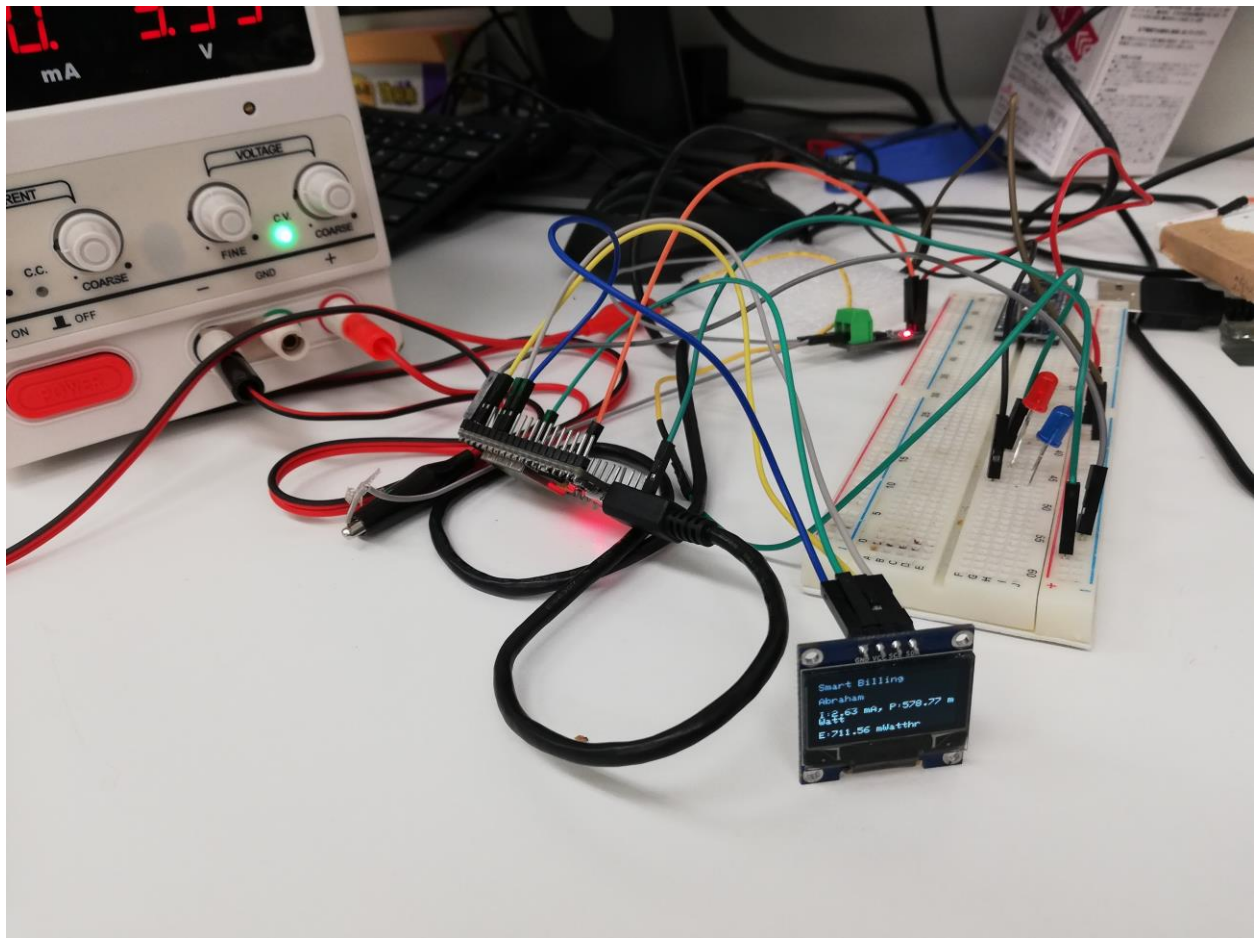
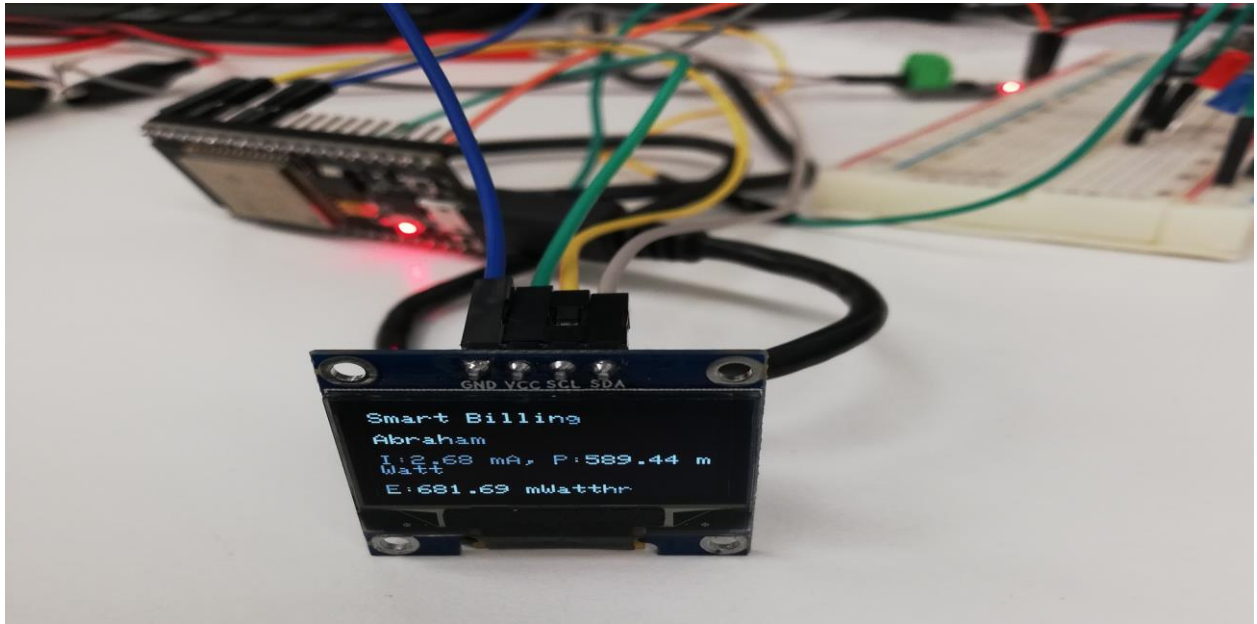


Figure L: Picture of Prototype experiment in the Lab

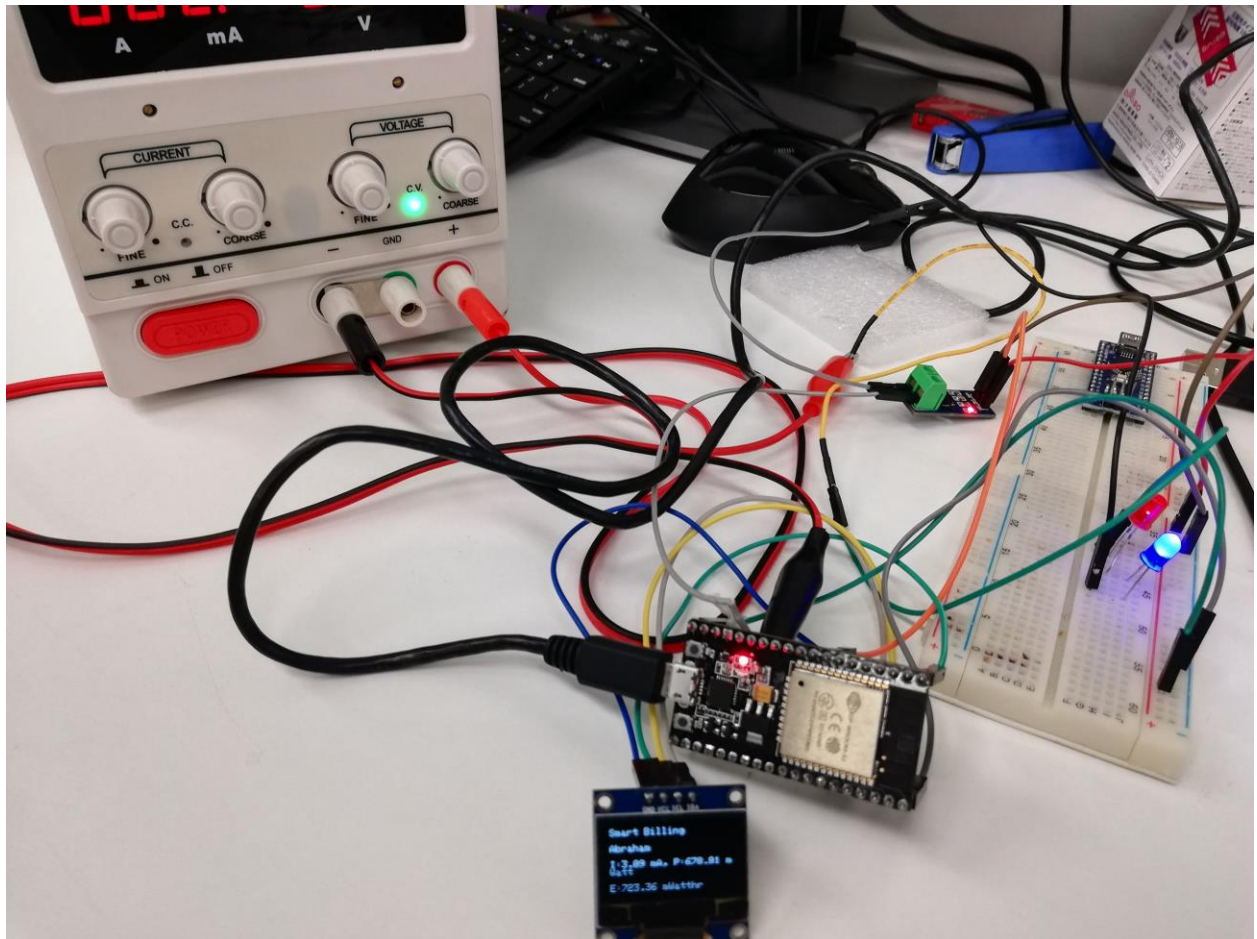


Figure M: Picture of Prototype experiment in the Lab

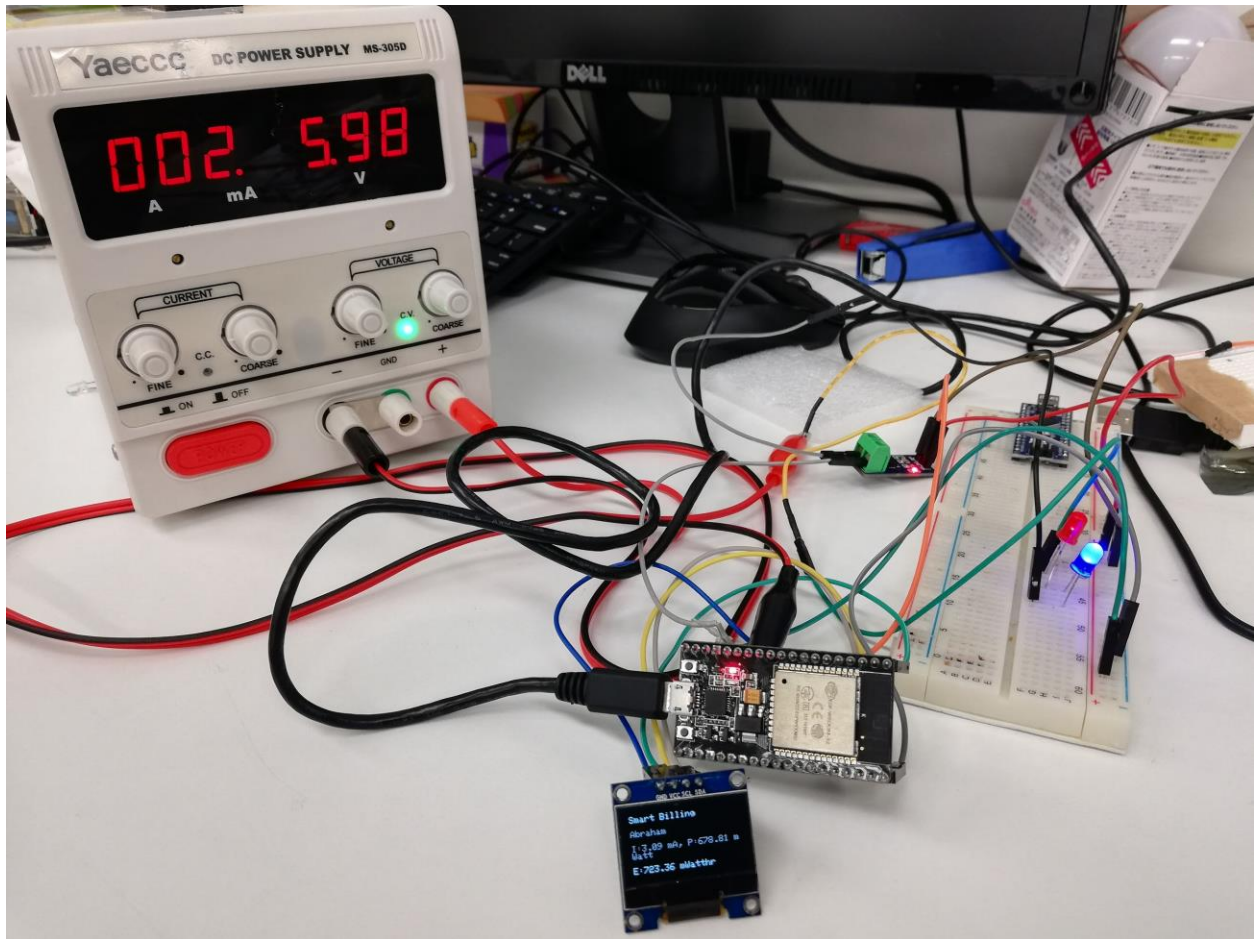


Figure N: Picture of Prototype experiment in the Lab